

Large increases in public R&D investment are needed to avoid declines of US agricultural productivity

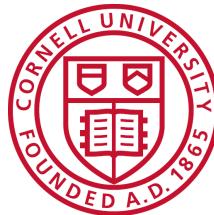
Ariel Ortiz-Bobea

Associate Professor of Applied Economics and Policy

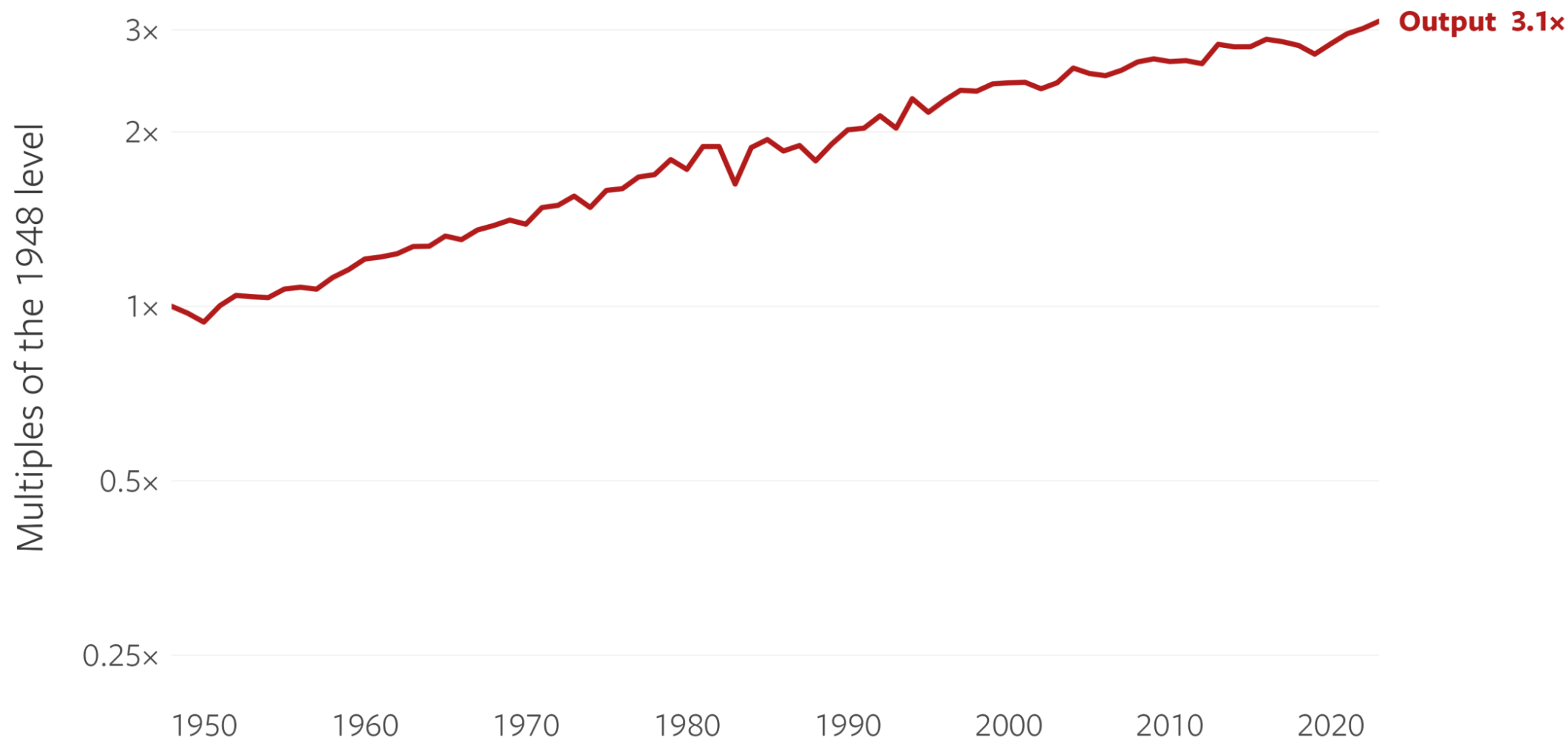
Dyson School of Applied Economics and Management · Brooks School of Public Policy · Cornell University

with Robert Chambers (Maryland), Yurou He (Cornell) and David Lobell (Stanford) · PNAS, 2025

agInnovation Fall Meeting · Grand Junction, Colorado · September 22, 2026

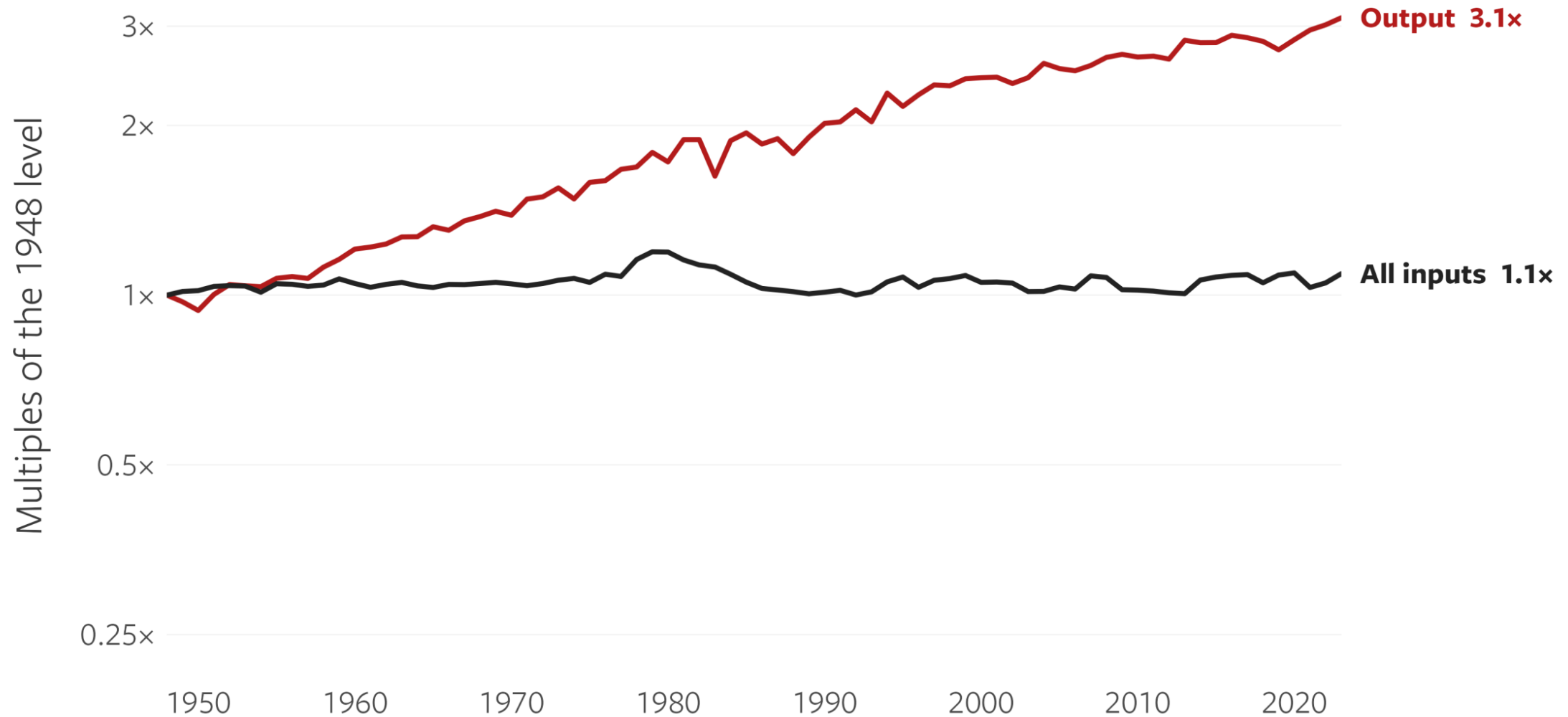


US agricultural output tripled since the 1950s



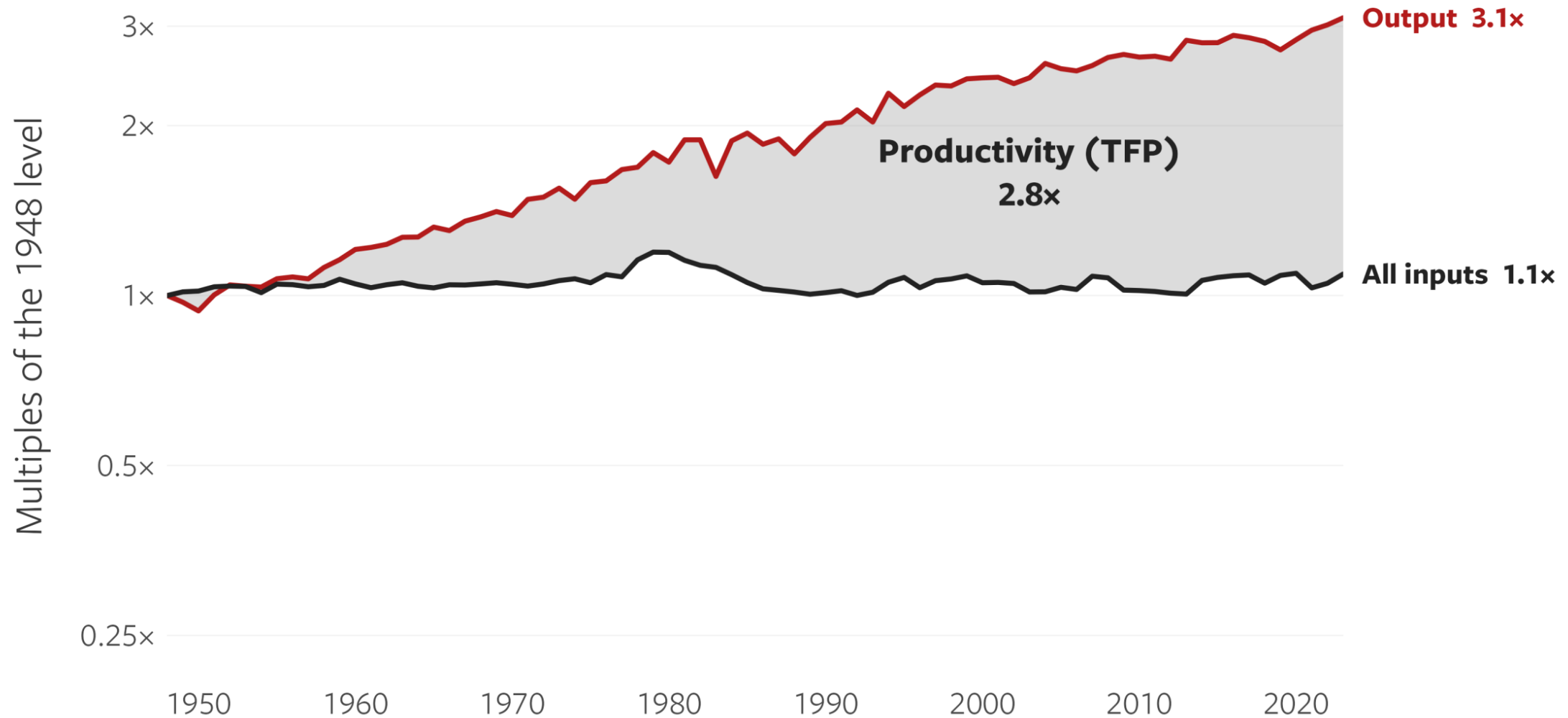
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

Output tripled with the same inputs



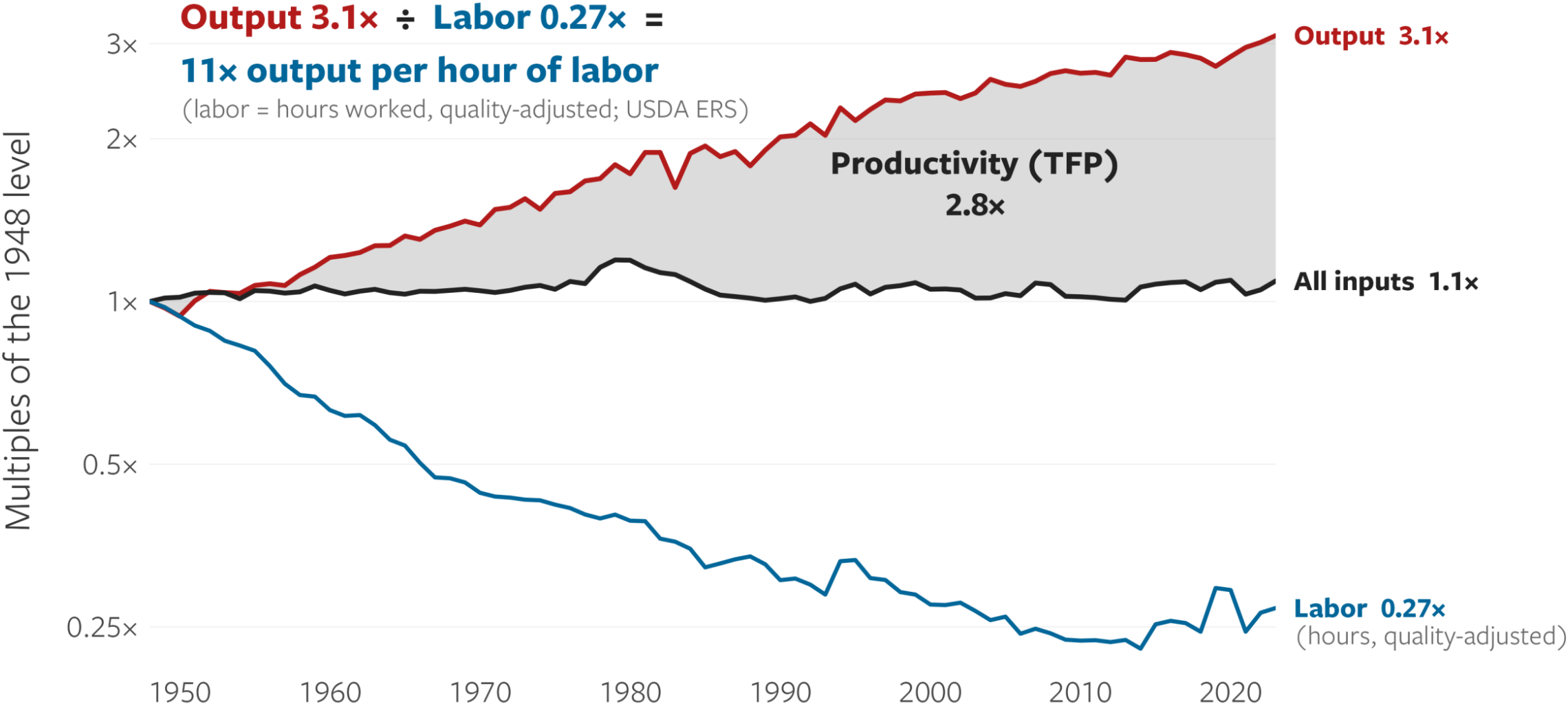
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

The gap between output and inputs is productivity



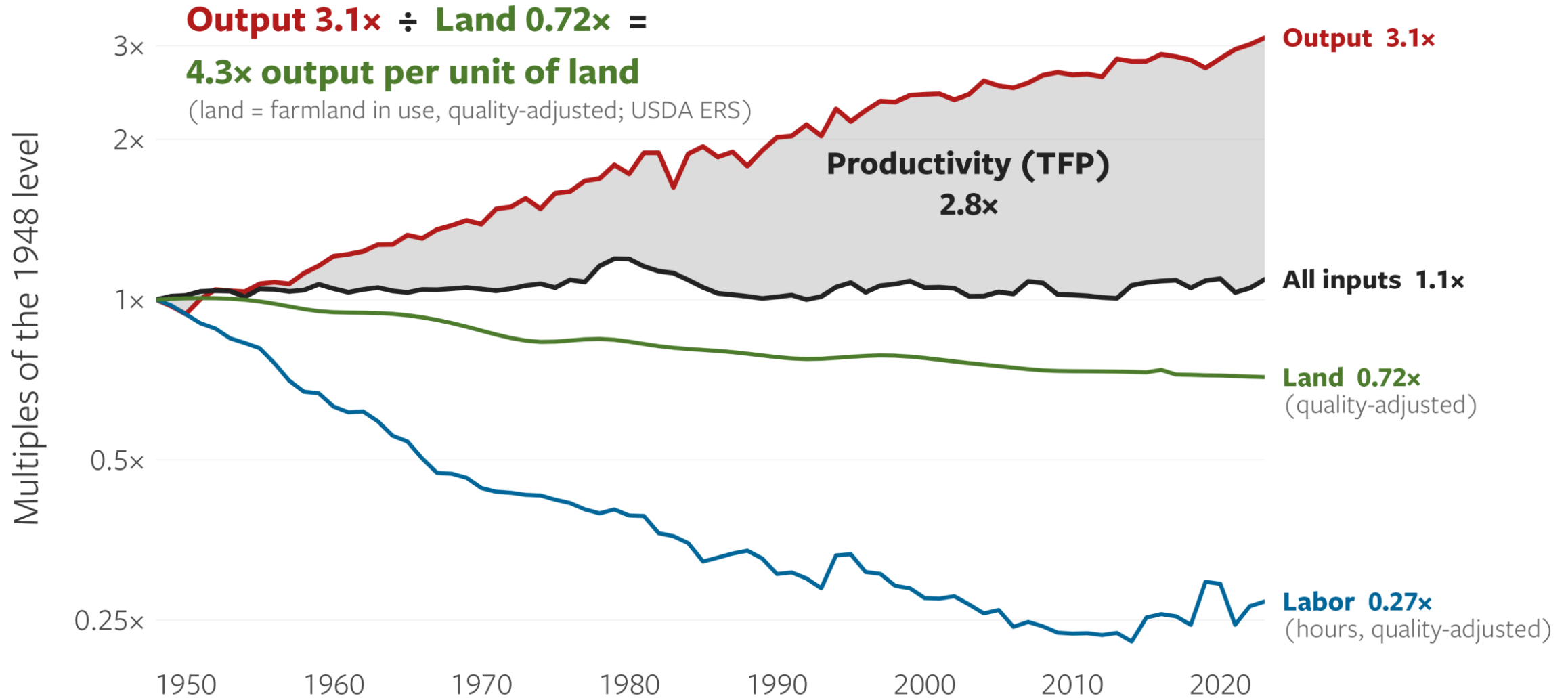
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

Good for farmers: 11x the output per hour of labor



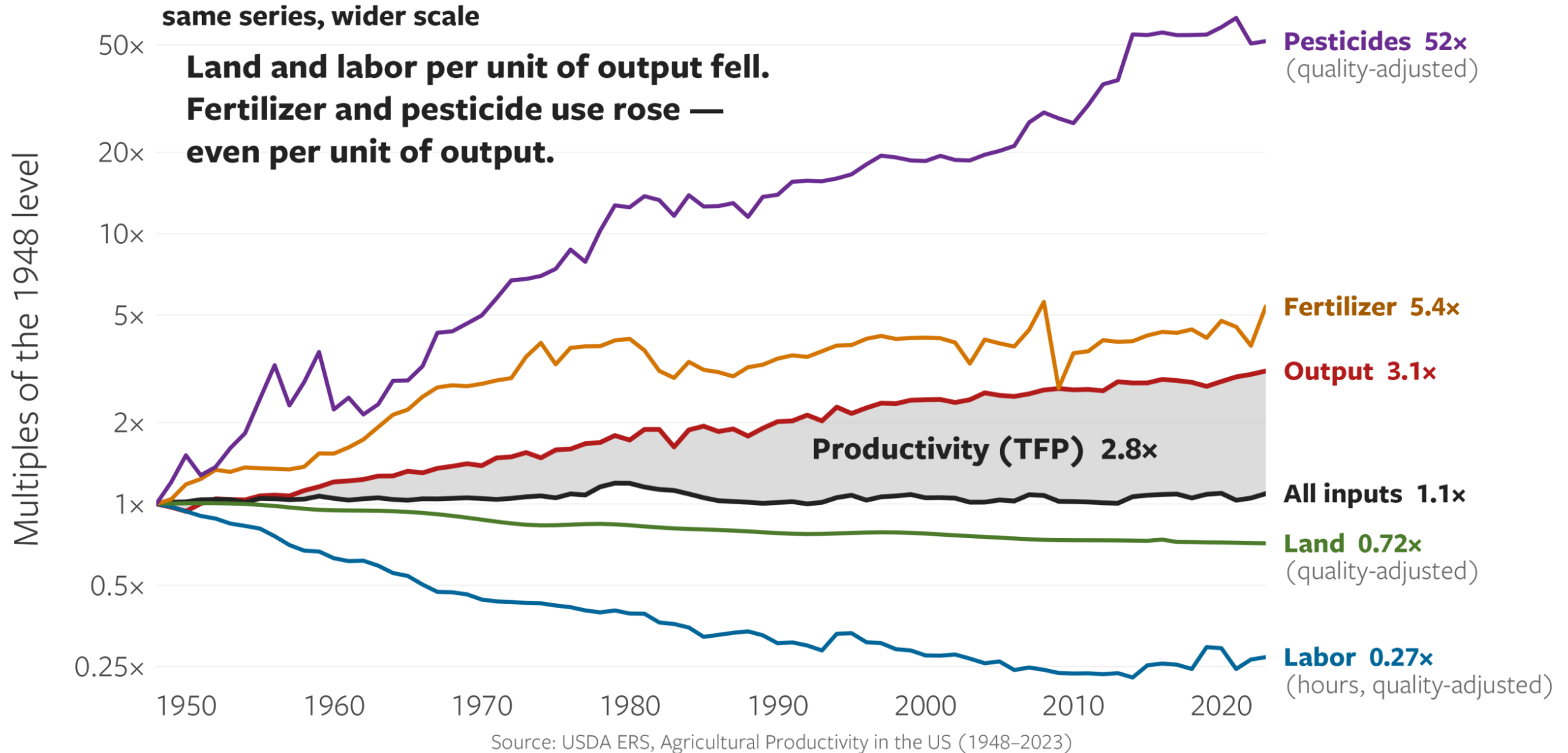
Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

Good for the environment: 4.3× the output per acre

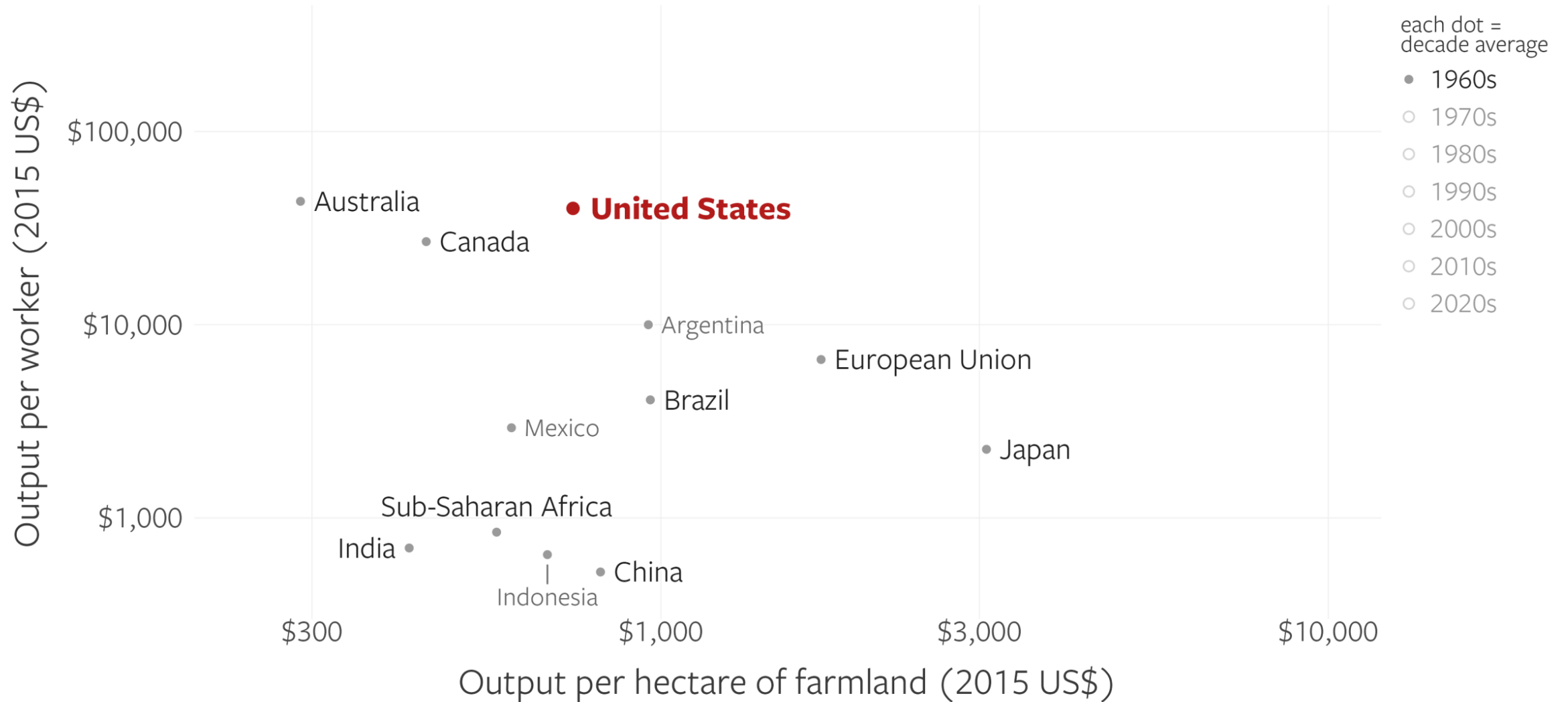


Source: USDA ERS, Agricultural Productivity in the US (1948–2023)

But fertilizer and pesticide use grew faster than output

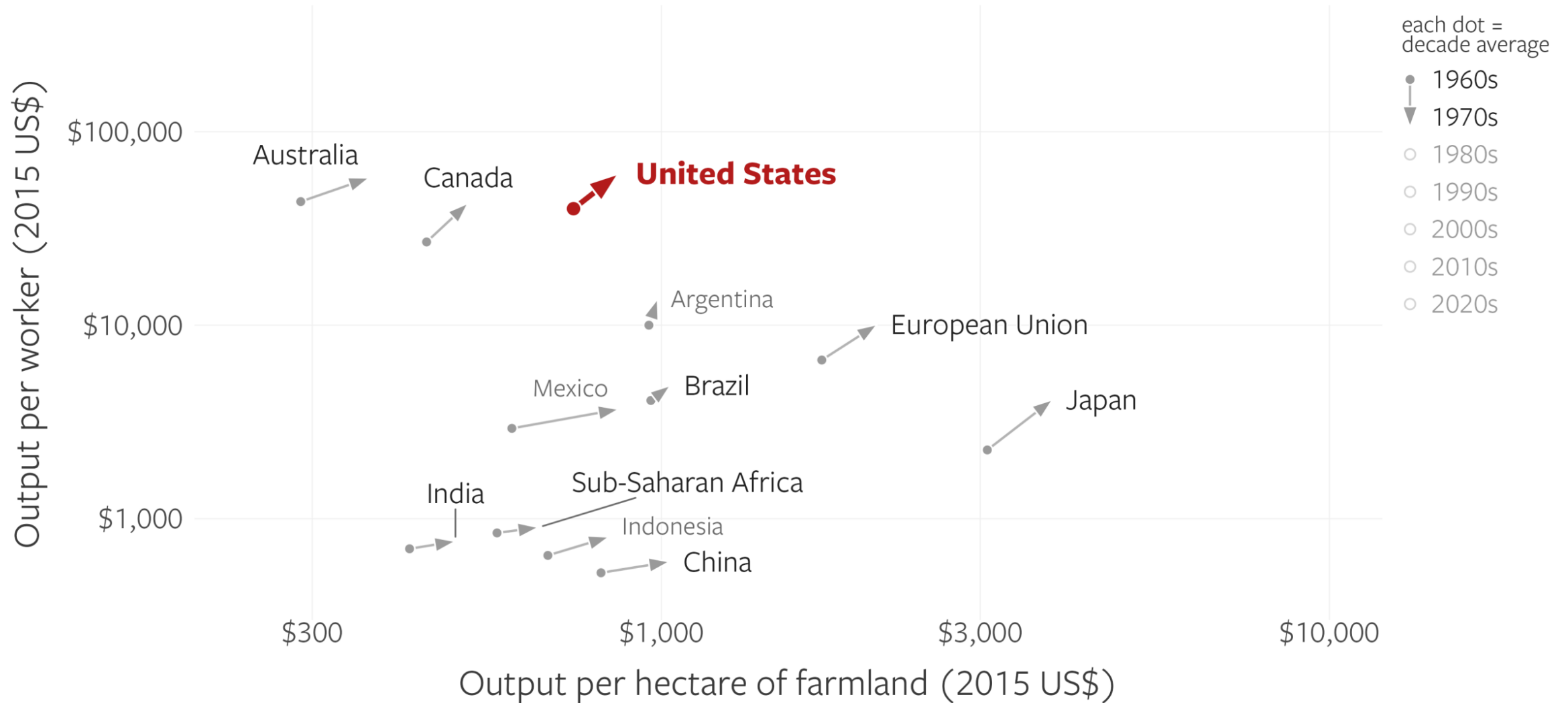


1960s



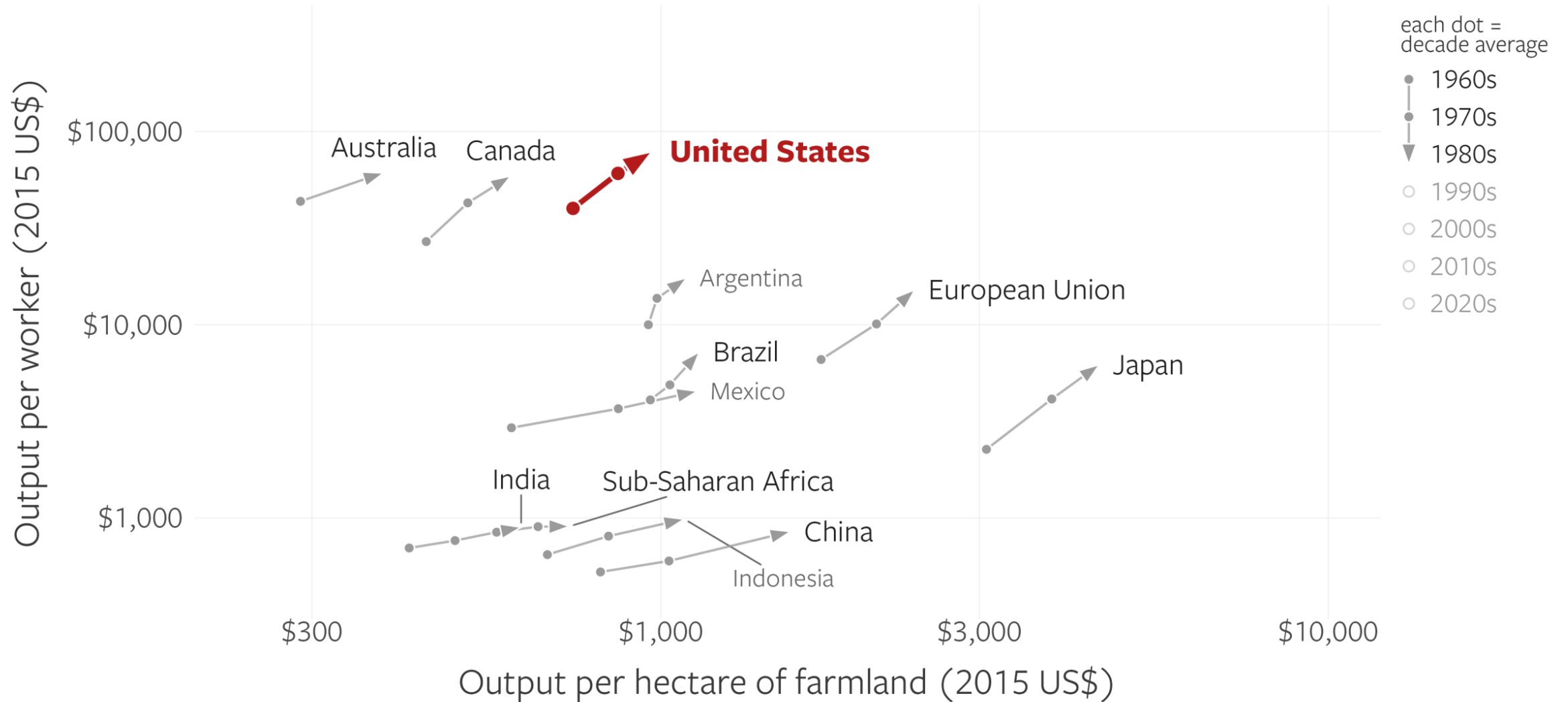
Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

1970s



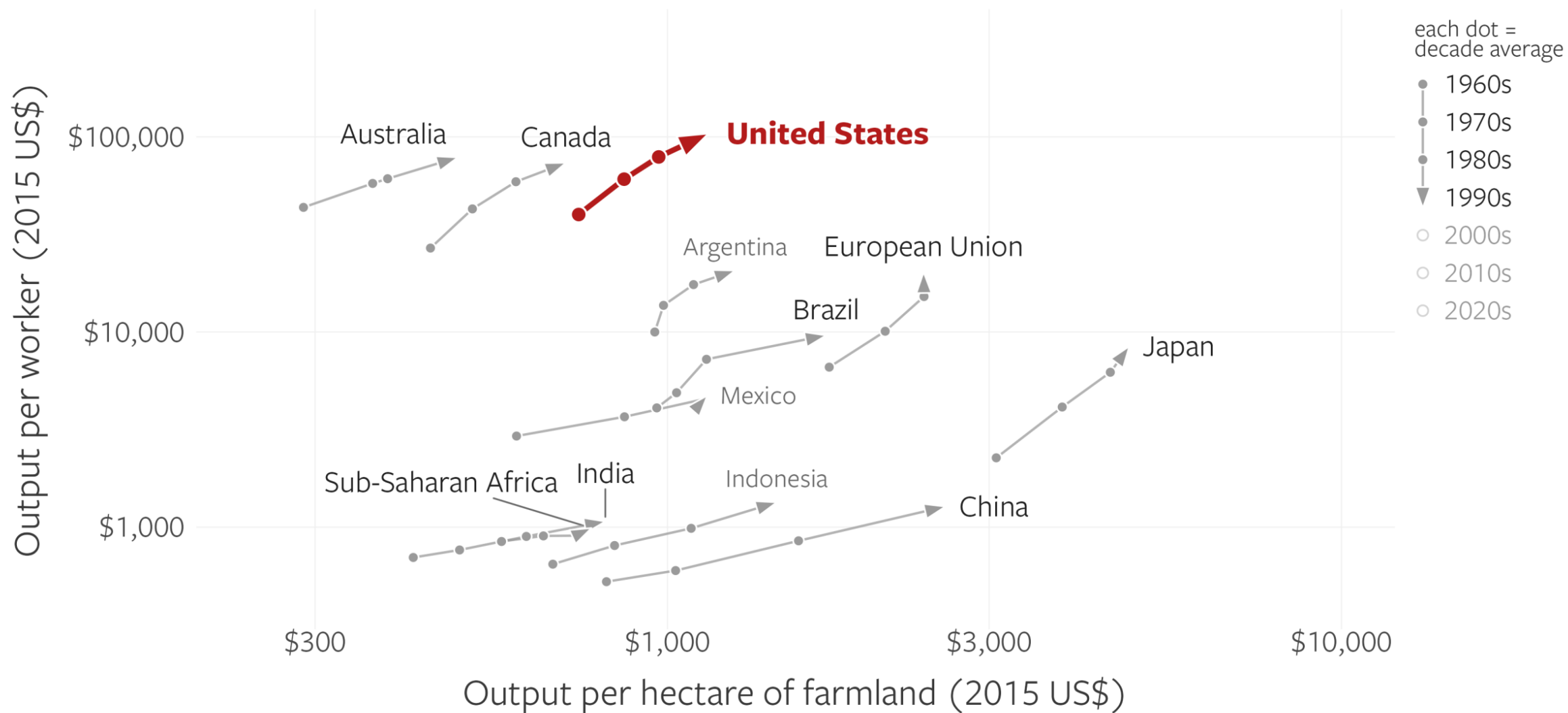
Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

1980s



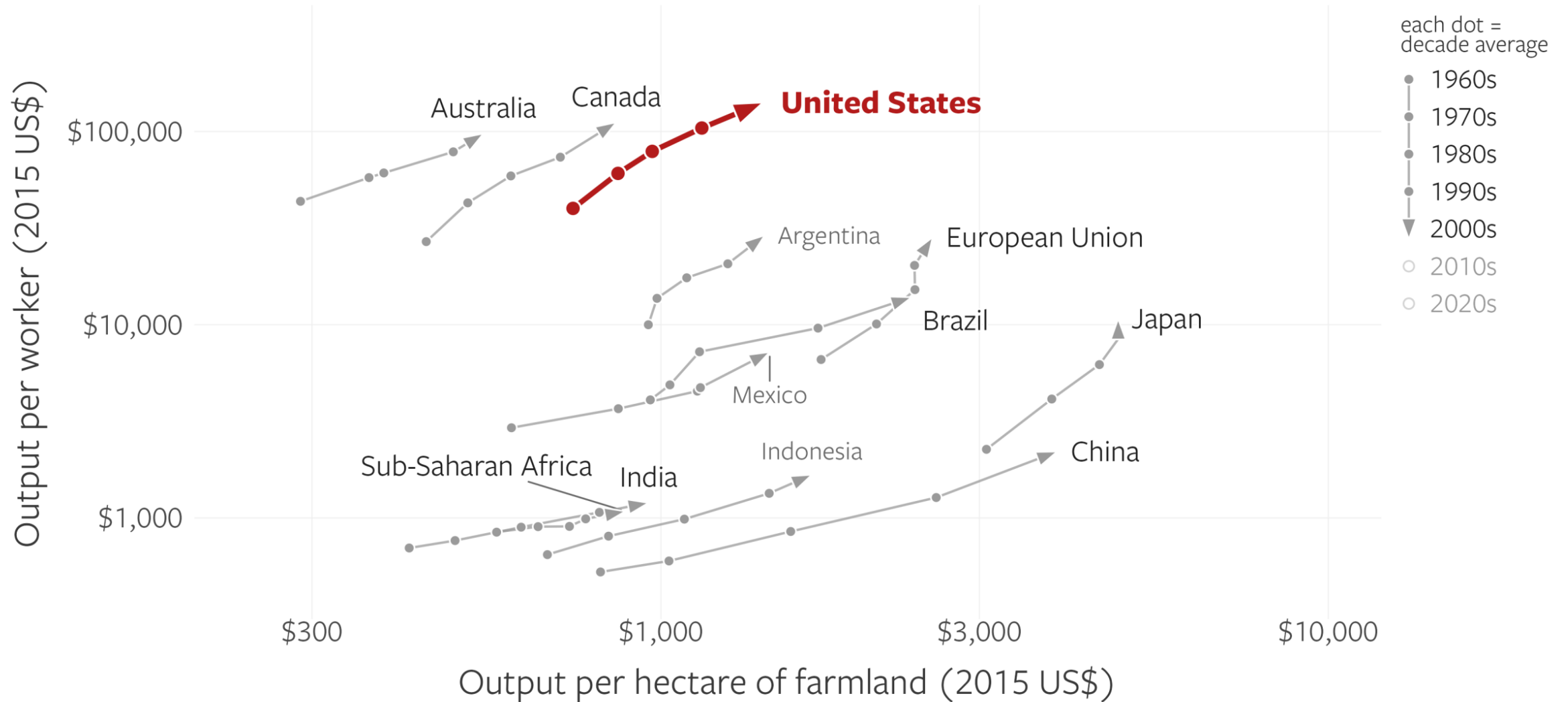
Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

1990s



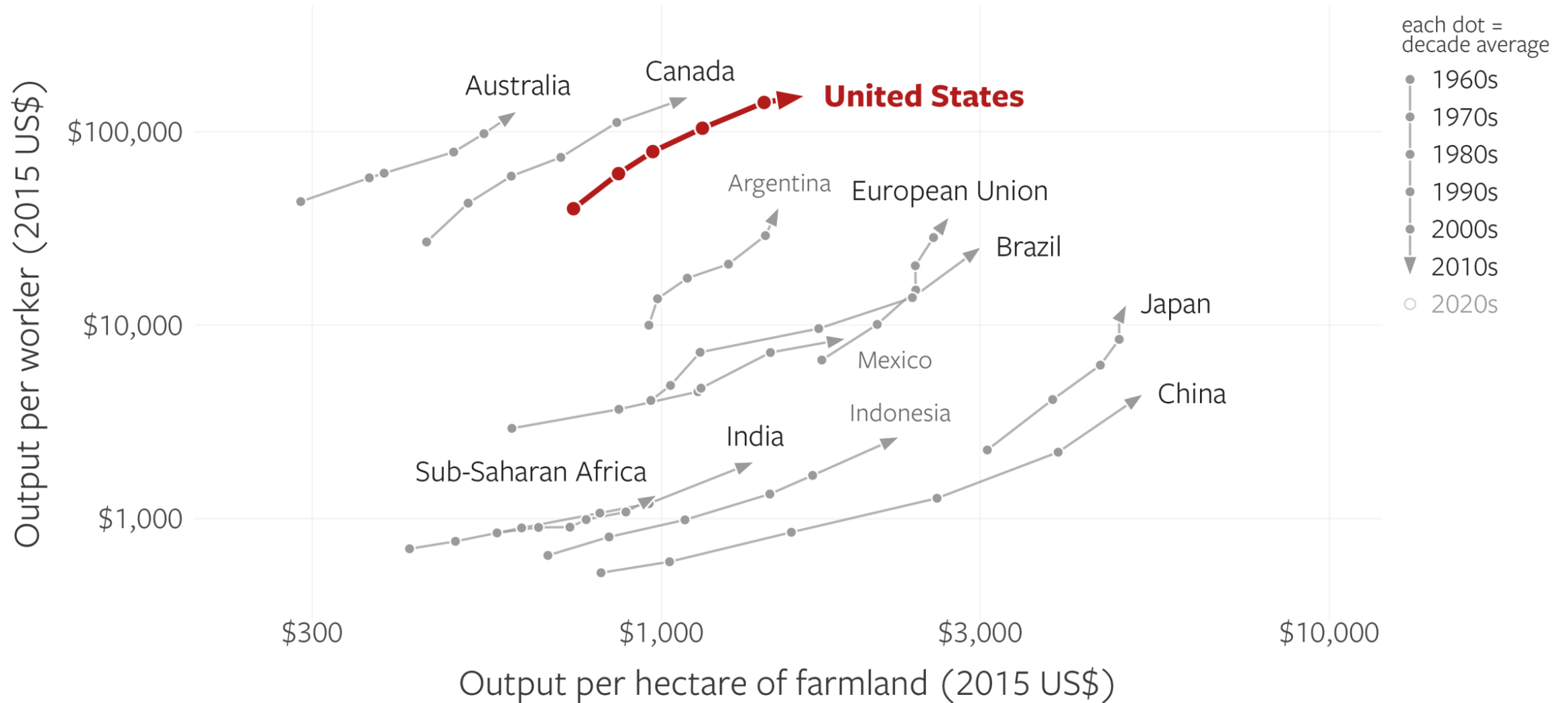
Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

2000s



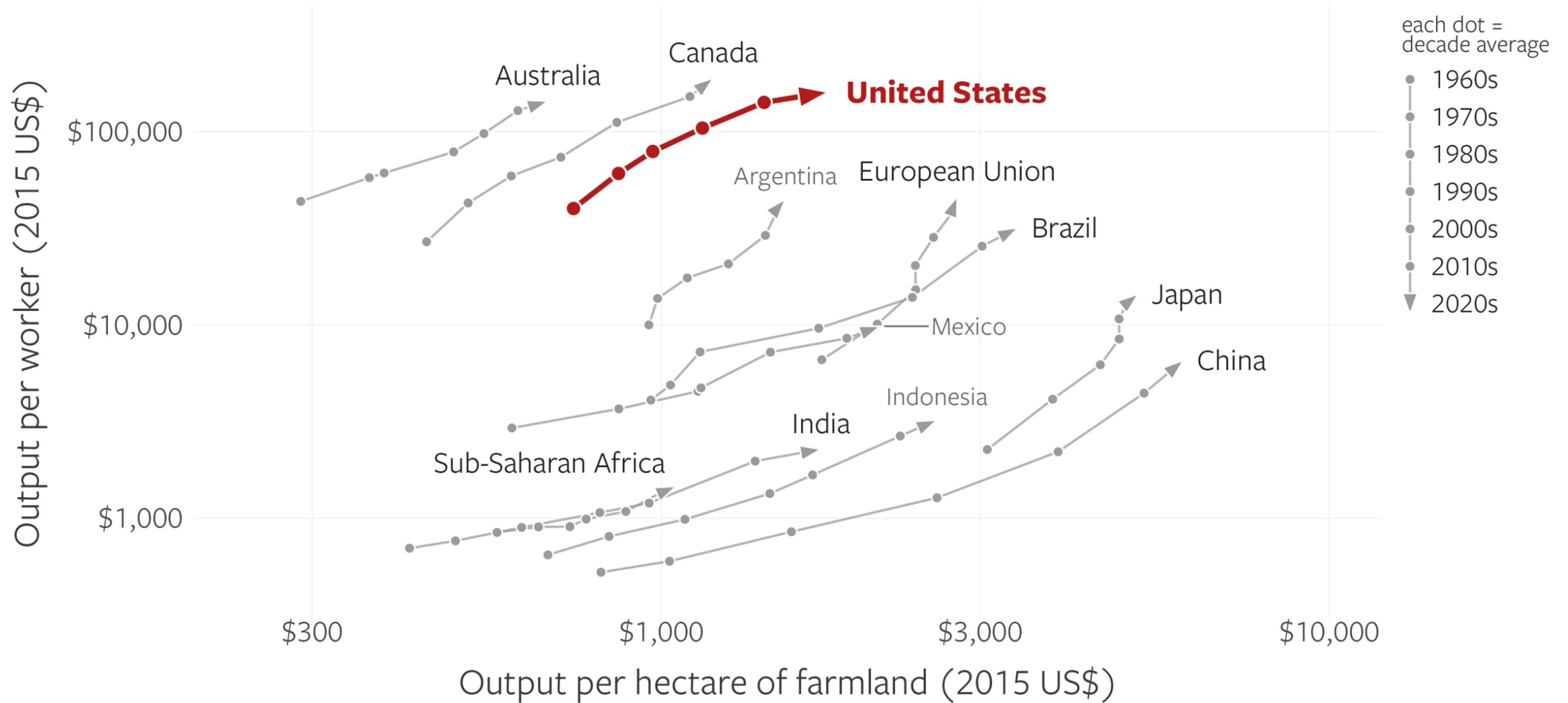
Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

2010s



Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

2020s



Source: USDA ERS, International Agricultural Productivity (1961–2023); each point is a decade average (2020s = 2020–23).

Key messages of this talk

1

Productivity growth is slowing.

2

**The main driver is slowing R&D —
but increasingly it's climate.**

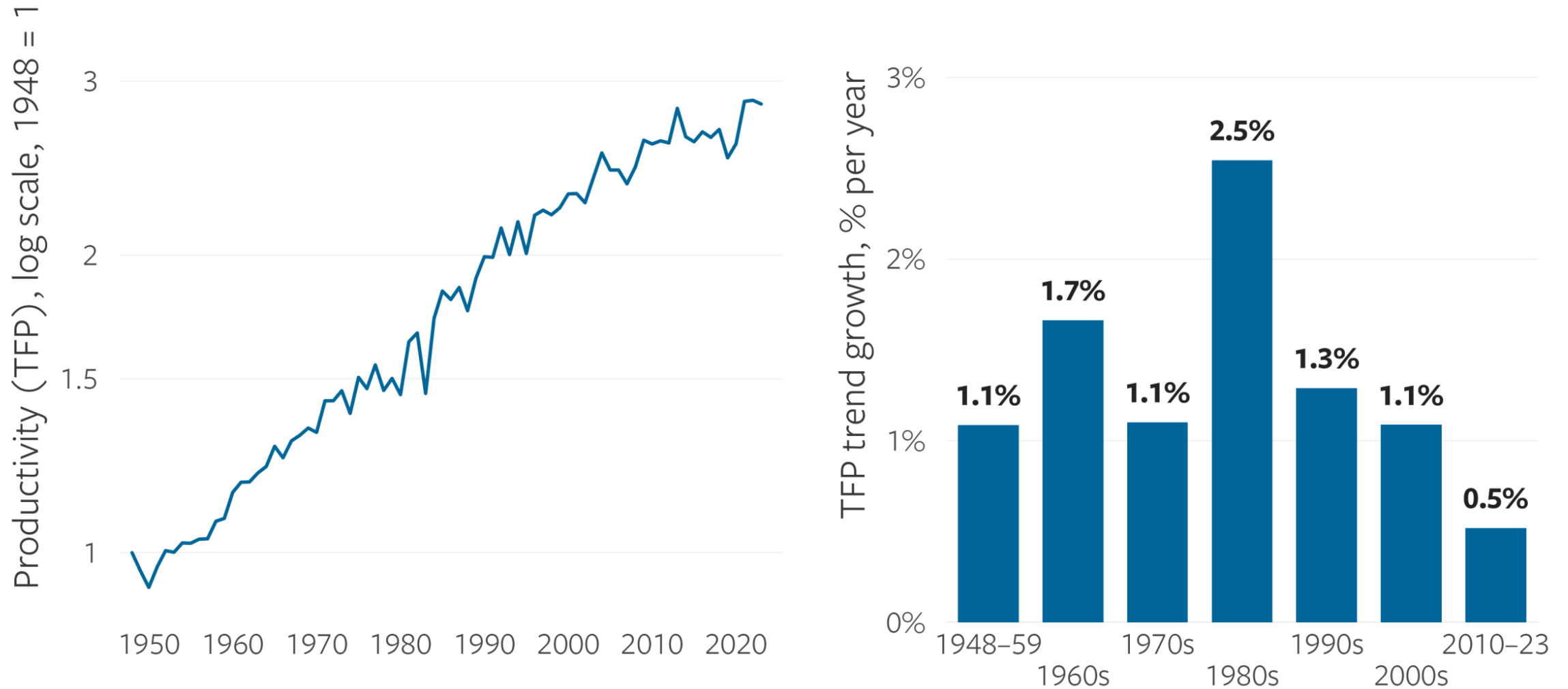
3

**Maintaining productivity growth
requires a historical pivot.**

Message 1

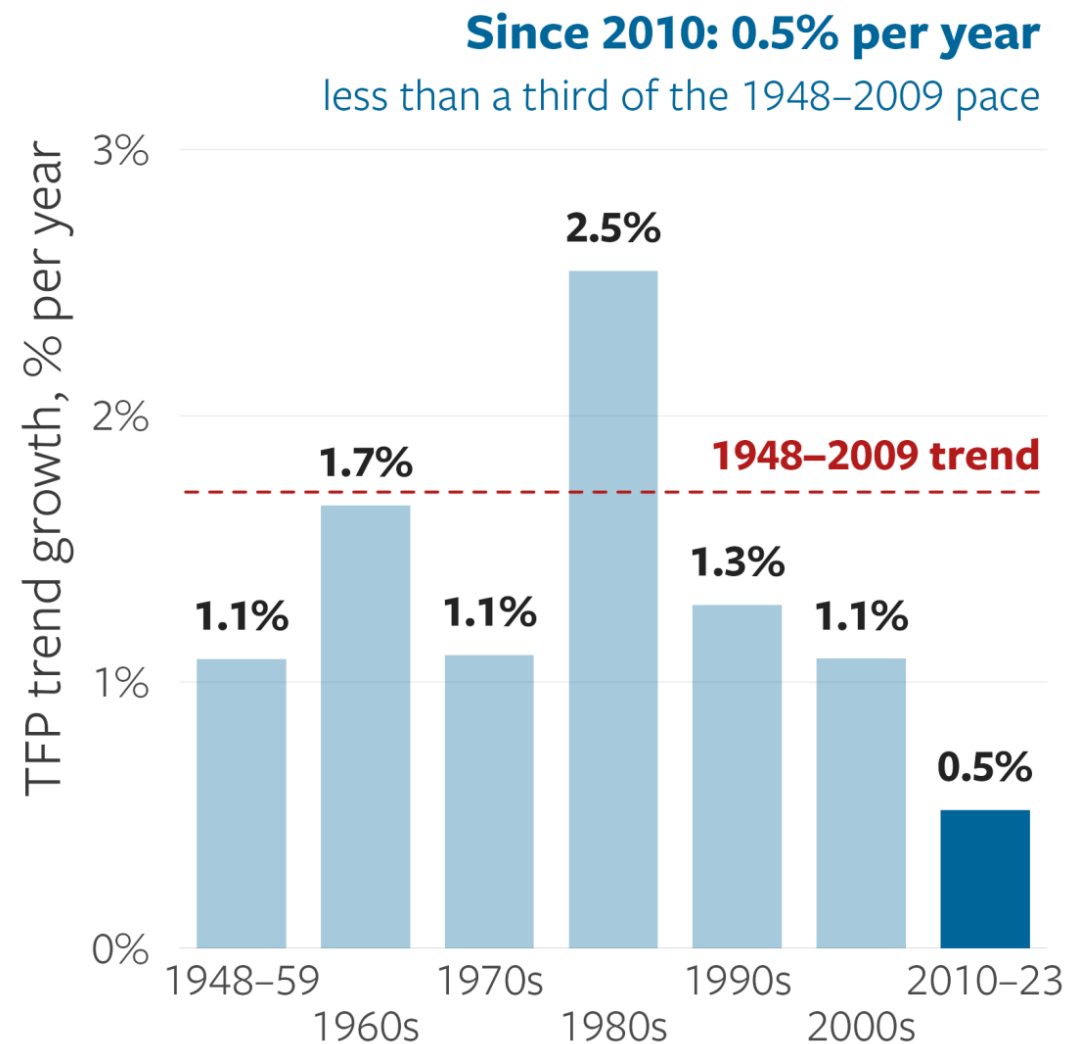
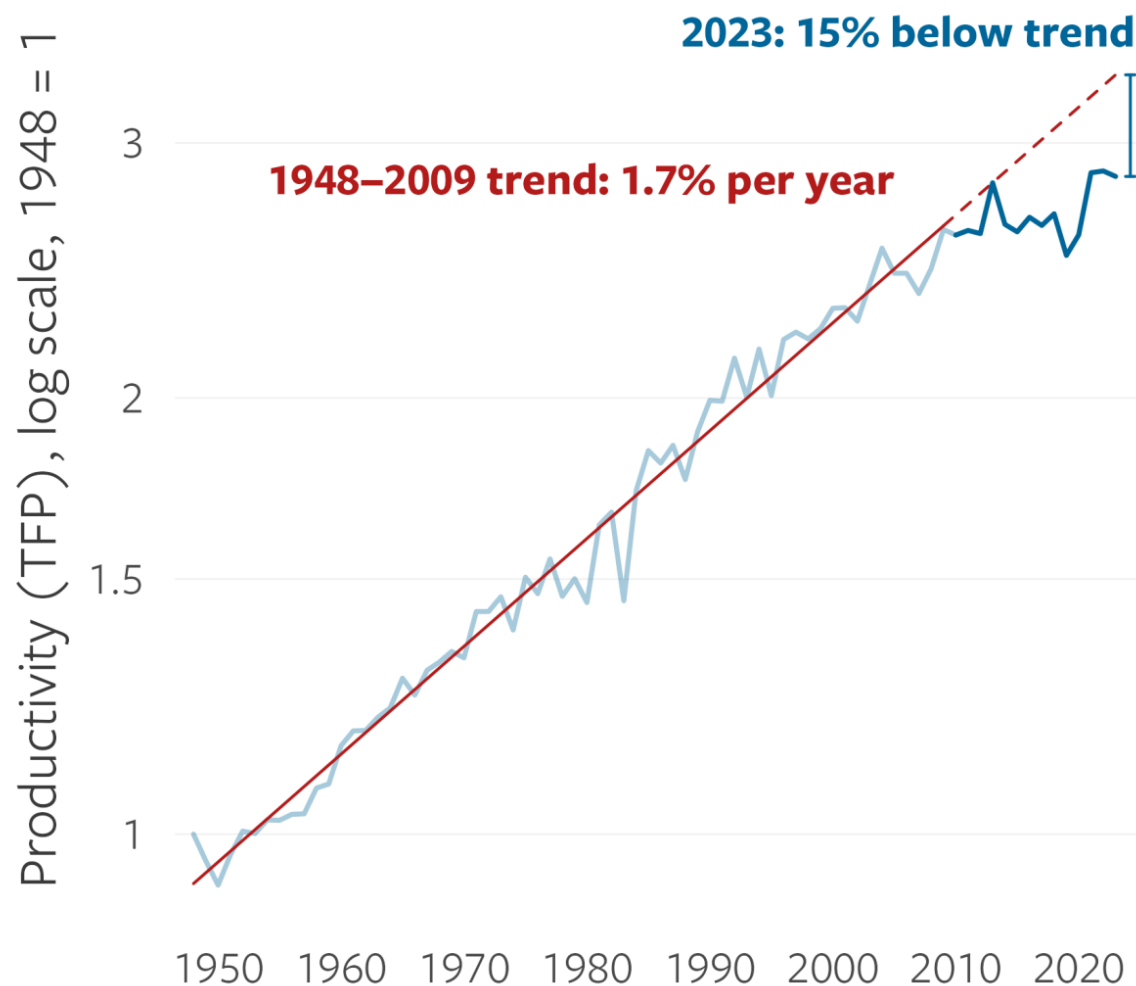
- 1 Productivity growth is slowing.**
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.**

Productivity growth, decade by decade



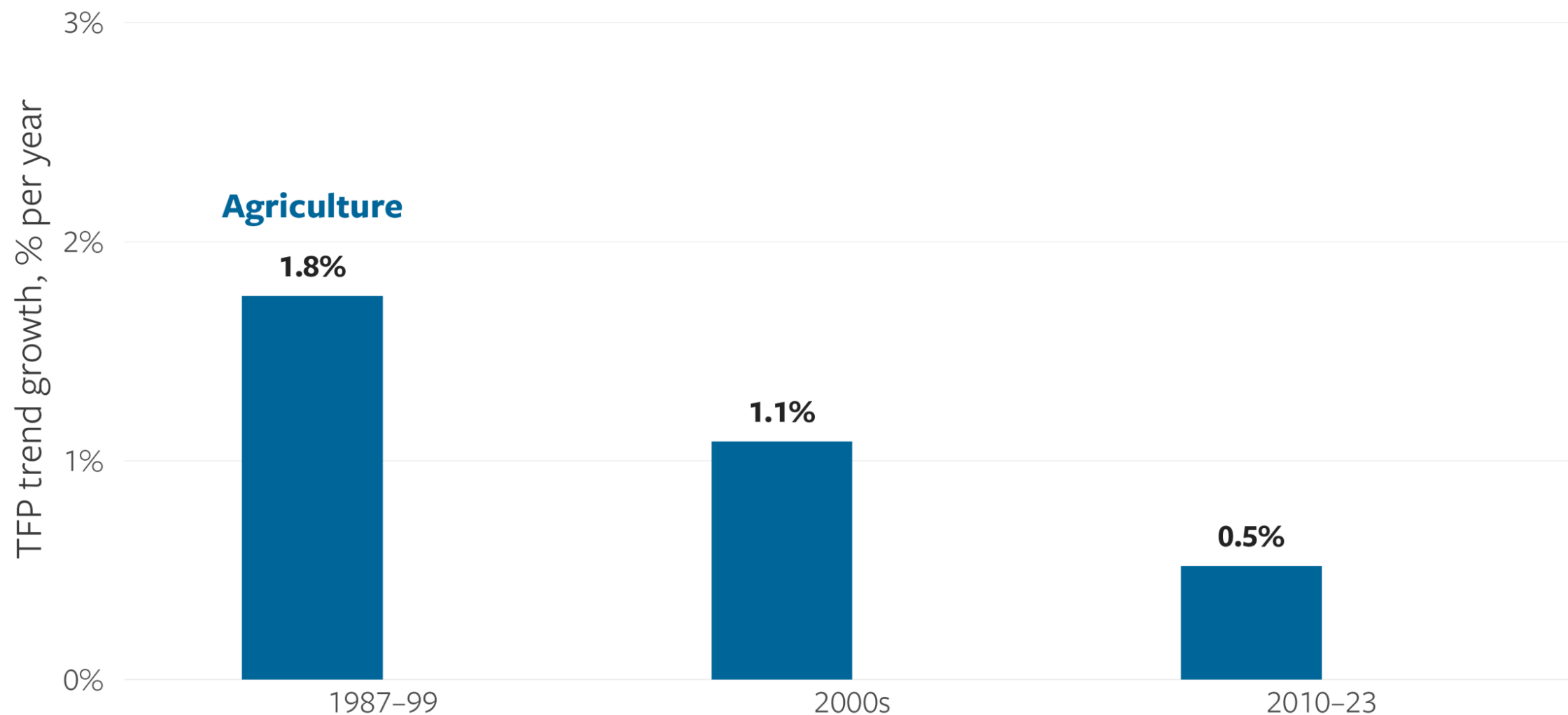
Source: USDA ERS, Agricultural Productivity in the US (1948-2023). Bars: trend growth of log TFP within each window.

The engine is slowing: productivity is 15% below its postwar trend



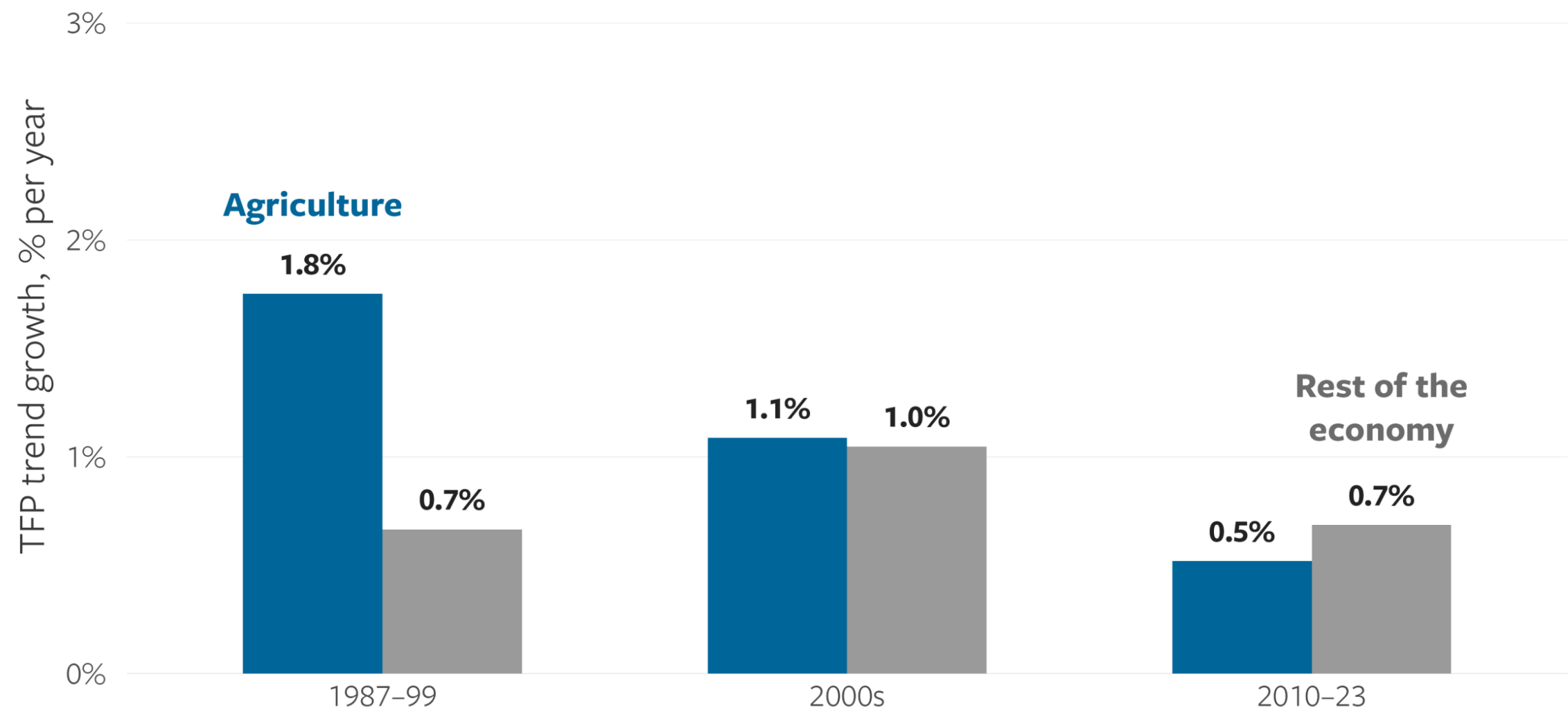
Source: USDA ERS, Agricultural Productivity in the US (1948–2023). Bars: trend growth of log TFP within each window.

Productivity growth by decade: agriculture



Source: USDA ERS (farm TFP); BLS (TFP, private nonfarm business sector). Trend growth within each window.

Agriculture led the economy for two decades; since 2010 it has fallen behind



Source: USDA ERS (farm TFP); BLS (TFP, private nonfarm business sector). Trend growth within each window.

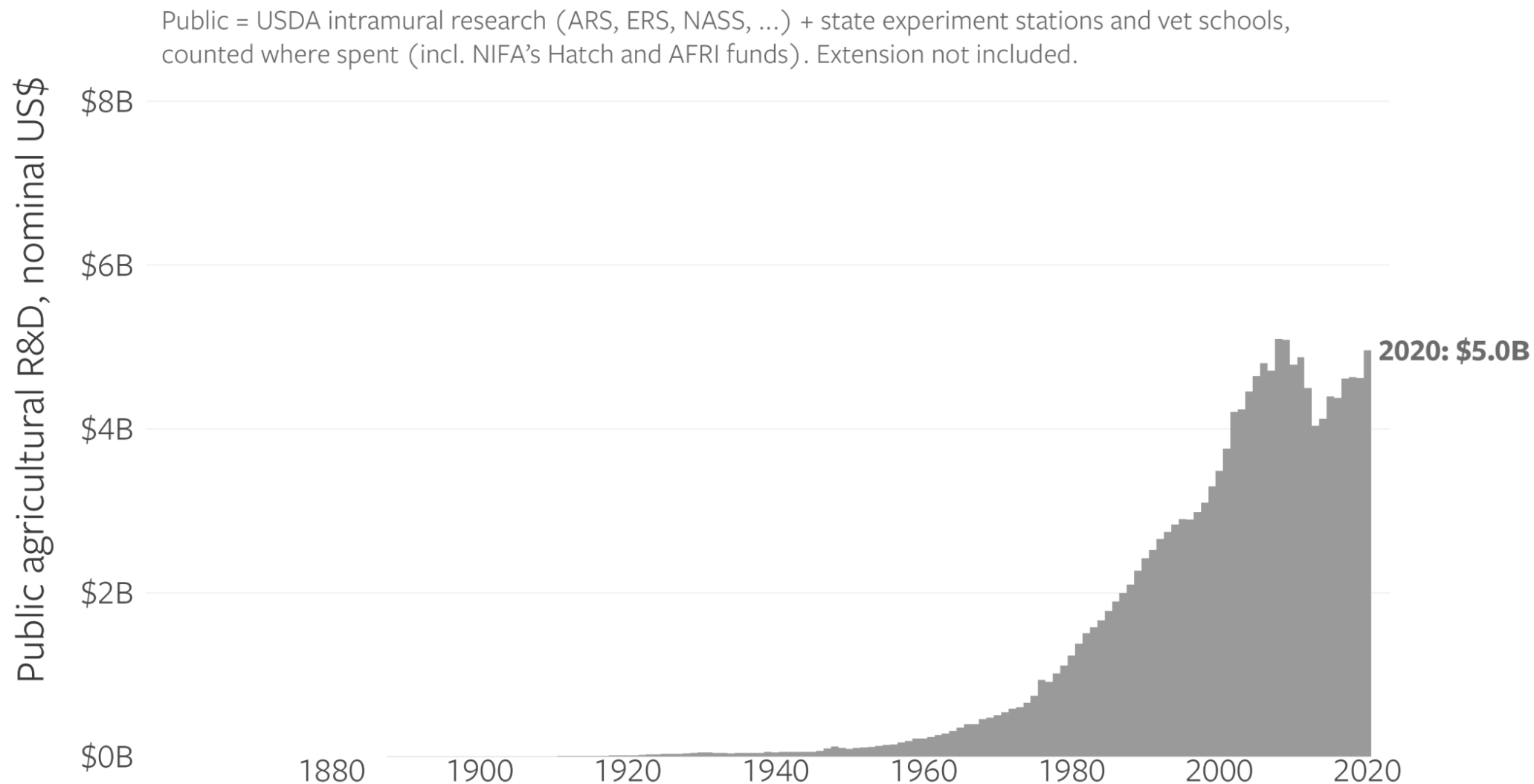
Message 1 — so why?

- 1 Productivity growth is slowing.**
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.**

Message 2

- 1 Productivity growth is slowing.
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.

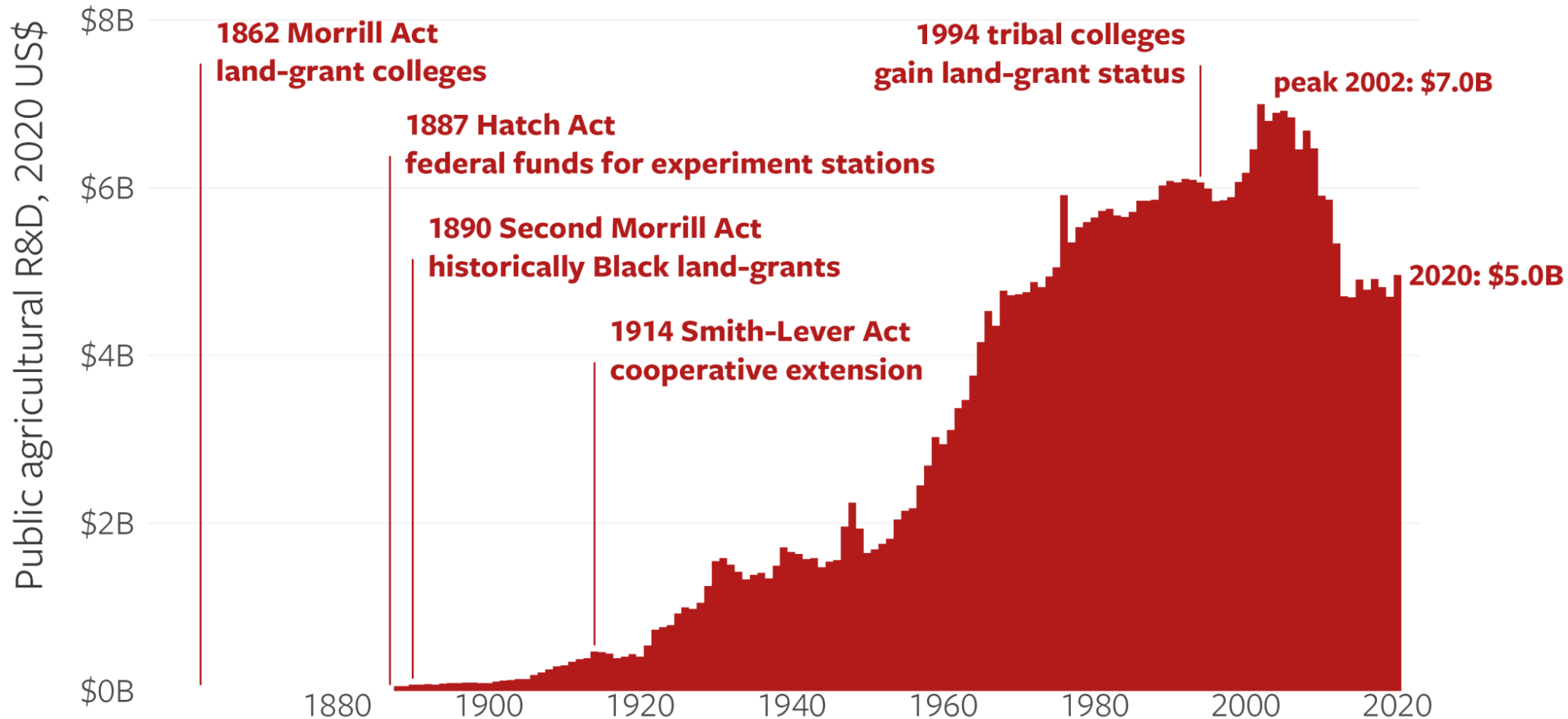
It looks like we spend more on research every year



Source: USDA ERS long-term public ag R&D series (Huffman & Evenson 2006 to 1950; ERS after), deflated with the NIH BRDPI; replication package of Ortiz-Bobea et al. (PNAS 2025).

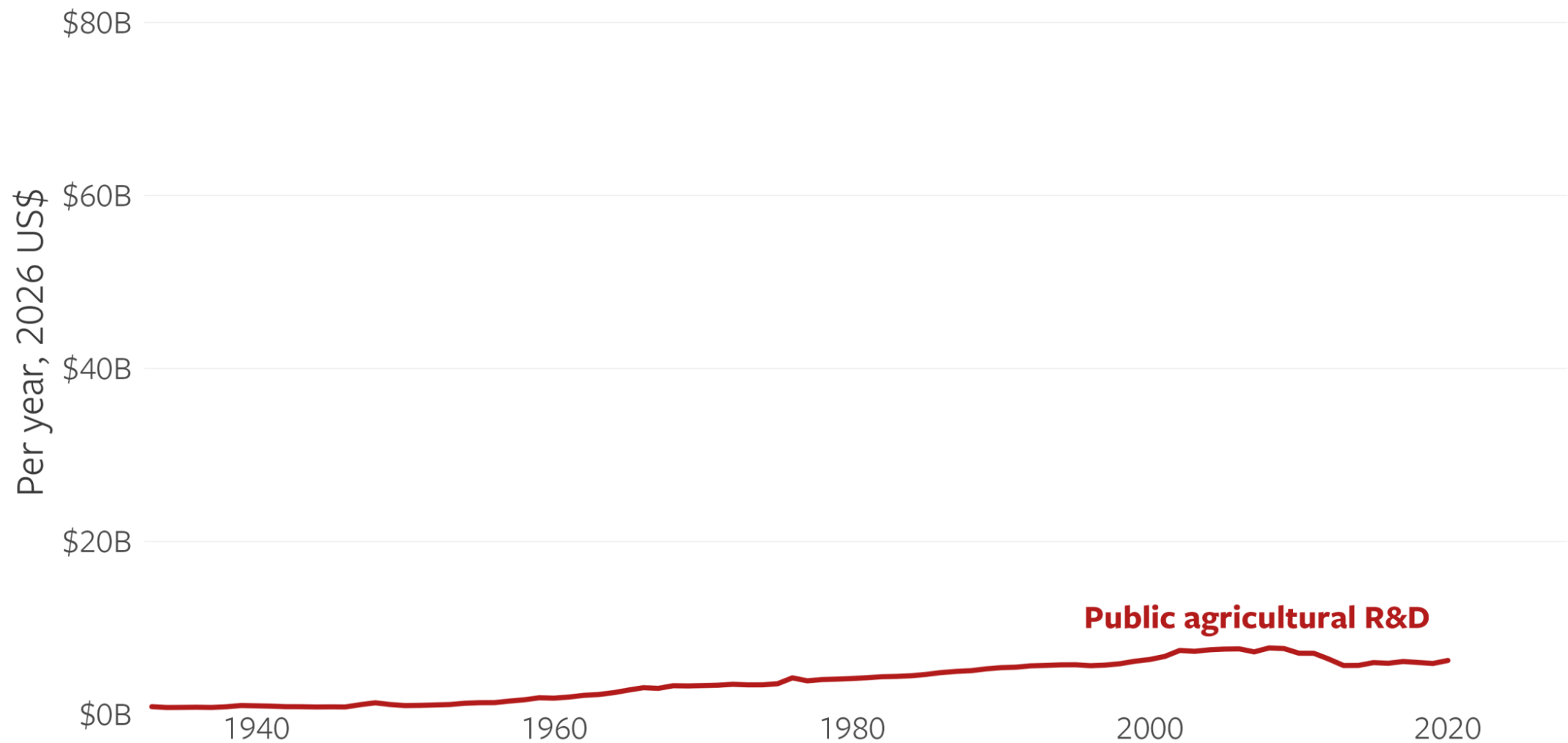
In real terms, we stopped forty years ago

Public = USDA intramural research (ARS, ERS, NASS, ...) + state experiment stations and vet schools, counted where spent (incl. NIFA's Hatch and AFRI funds). Extension not included.



Source: USDA ERS long-term public ag R&D series (Huffman & Evenson 2006 to 1950; ERS after), deflated with the NIH BRDPI; replication package of Ortiz-Bobea et al. (PNAS 2025).

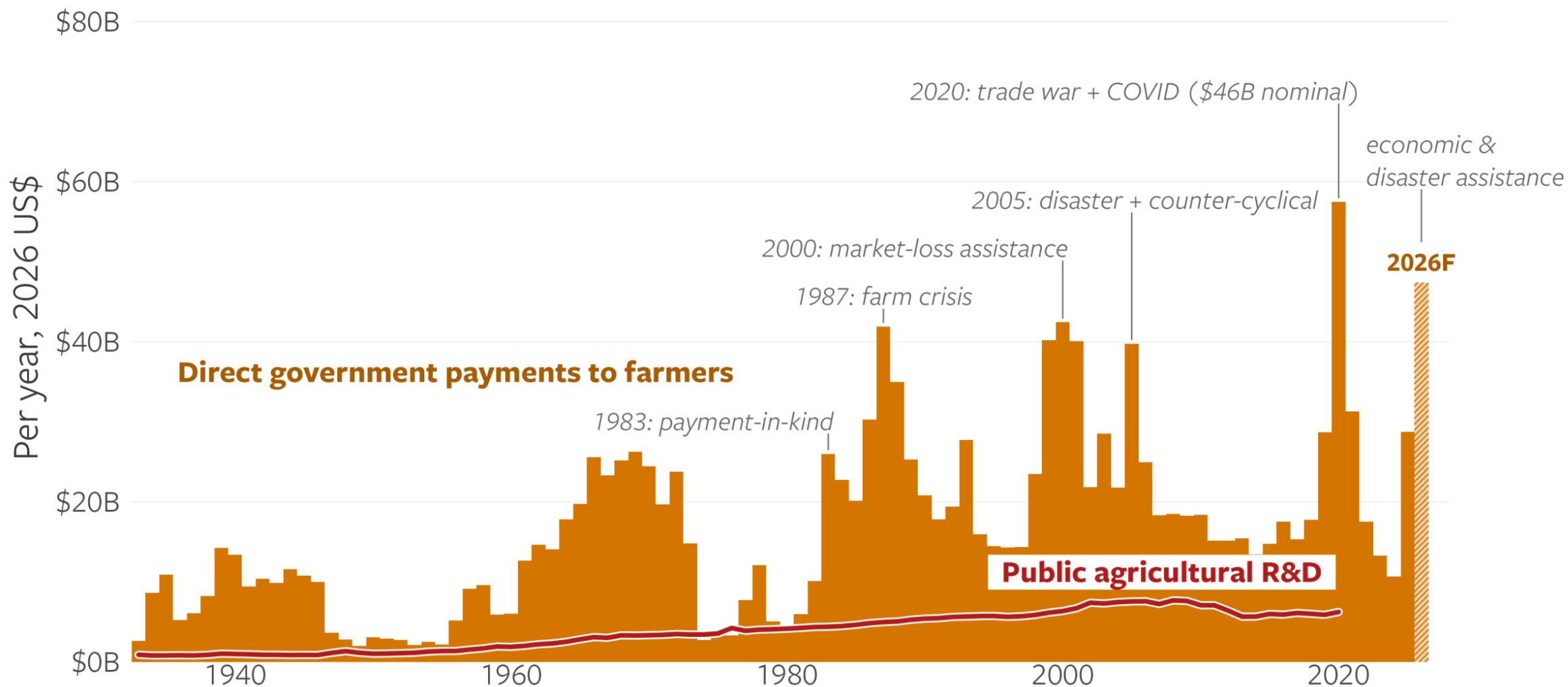
Public money in agriculture, in today's dollars: research...



Public agricultural R&D

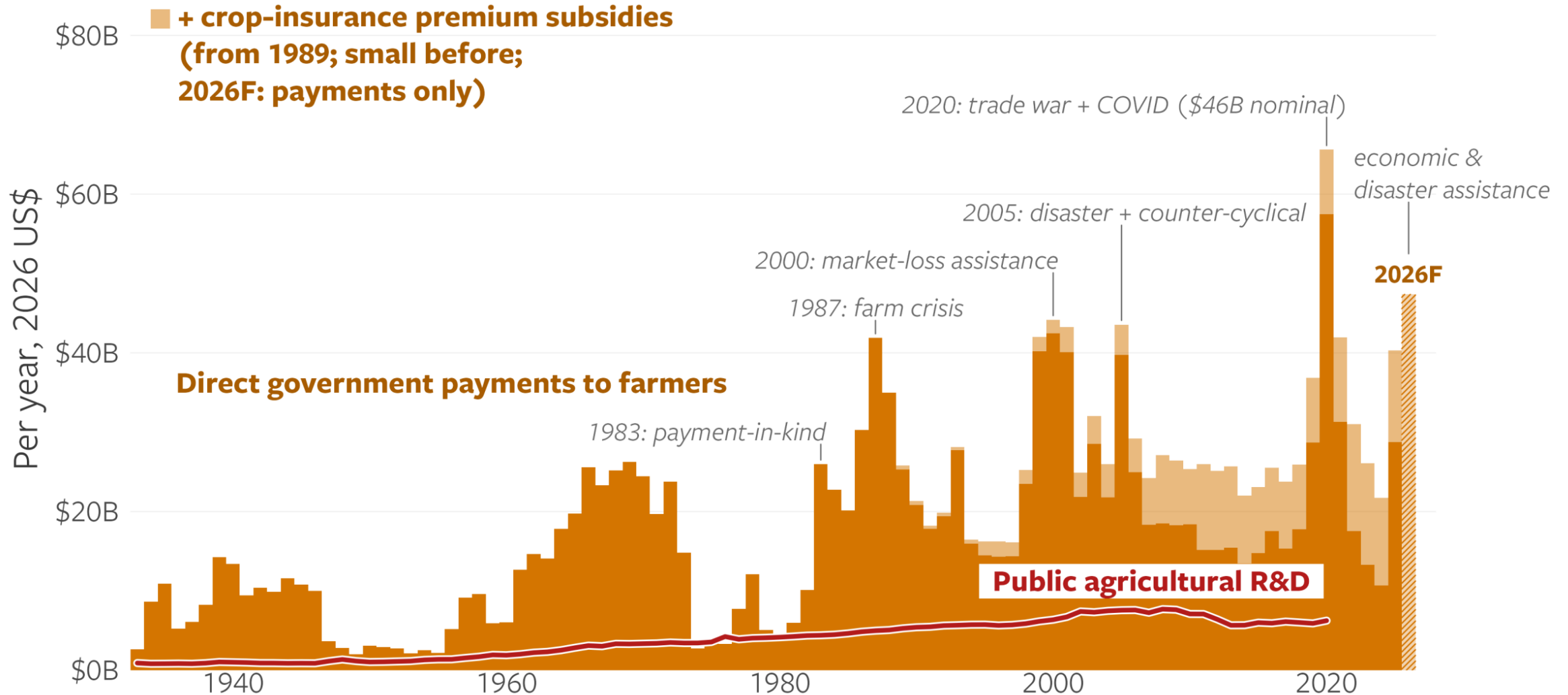
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D. All in 2026 \$ via the GDP deflator.

...and direct payments to farmers



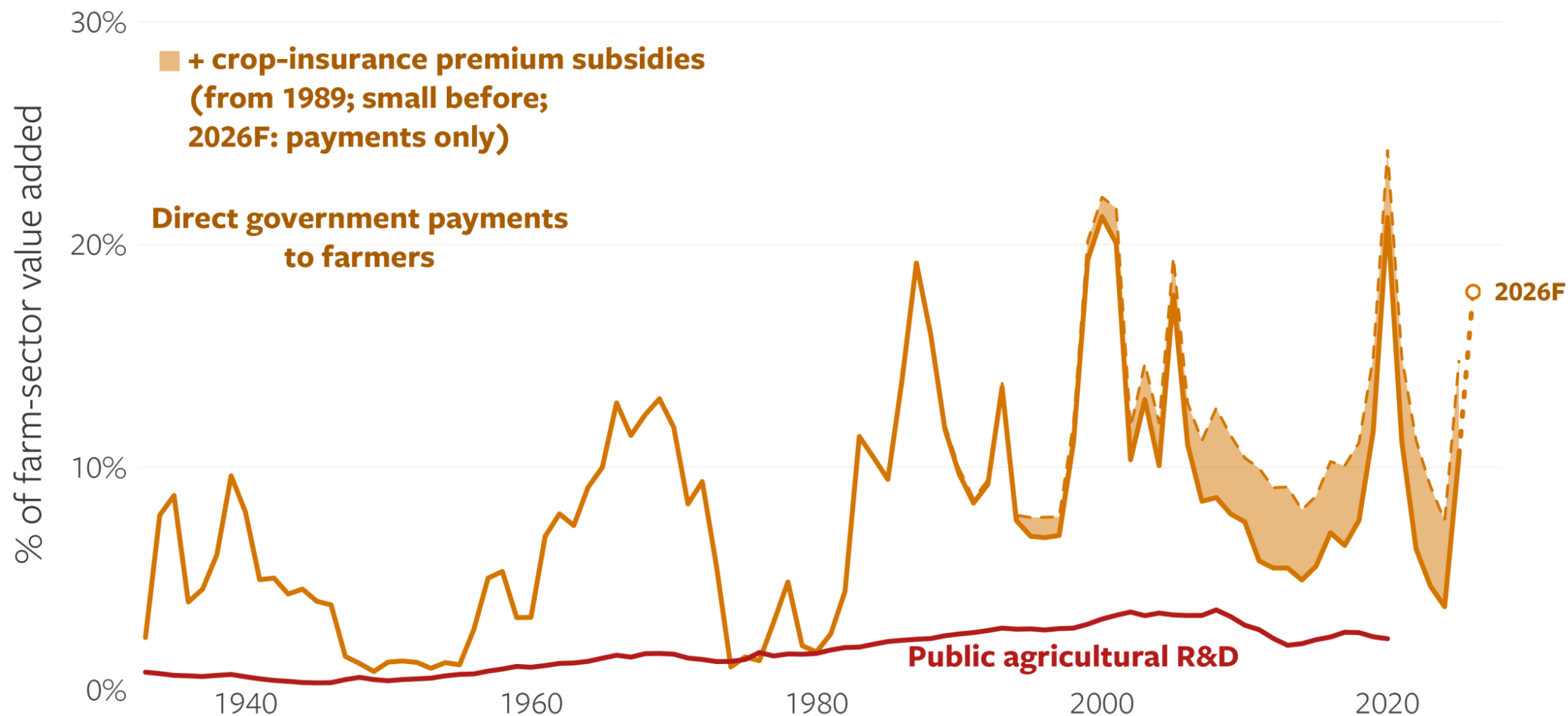
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D. All in 2026 \$ via the GDP deflator.

...and crop-insurance premium subsidies



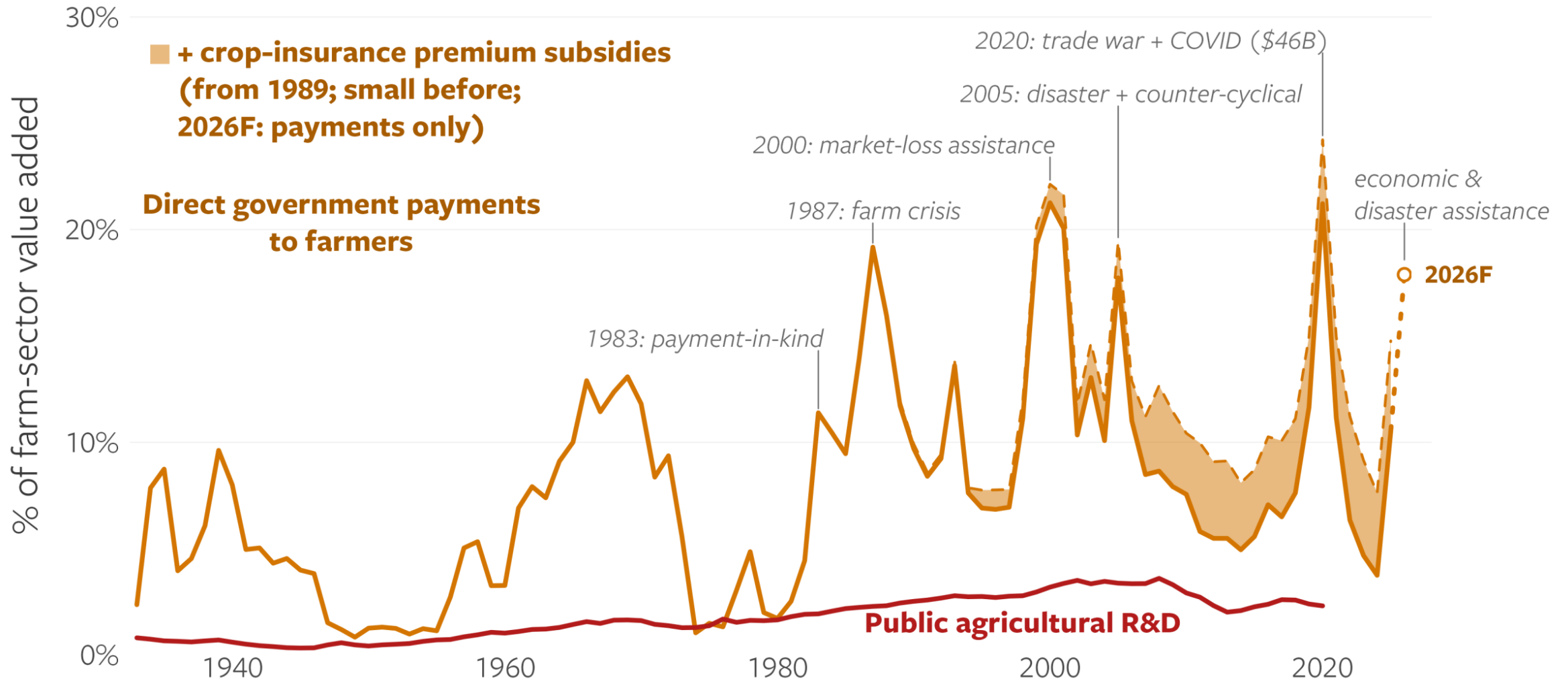
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D. All in 2026 \$ via the GDP deflator.

As a share of farm GDP: research versus farm support



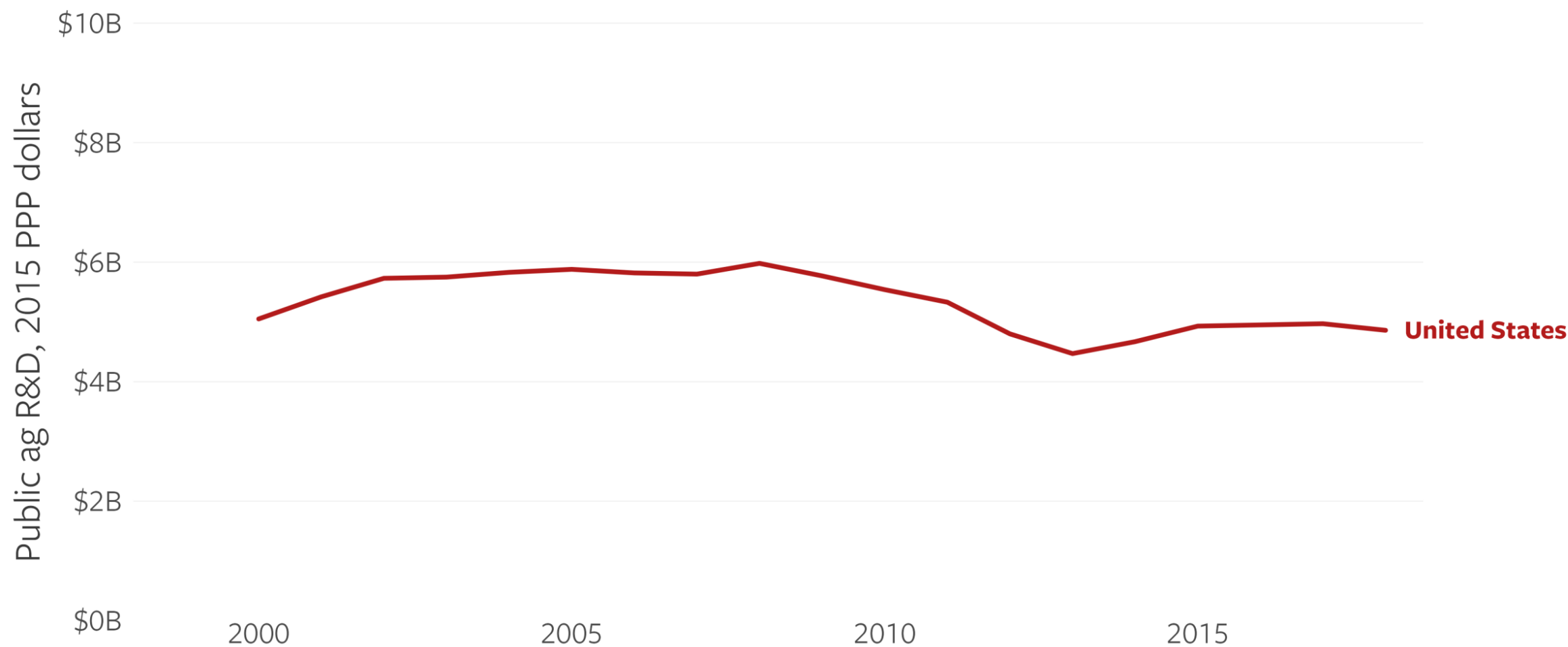
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026 release; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D.

Both are legitimate. Only one compounds.



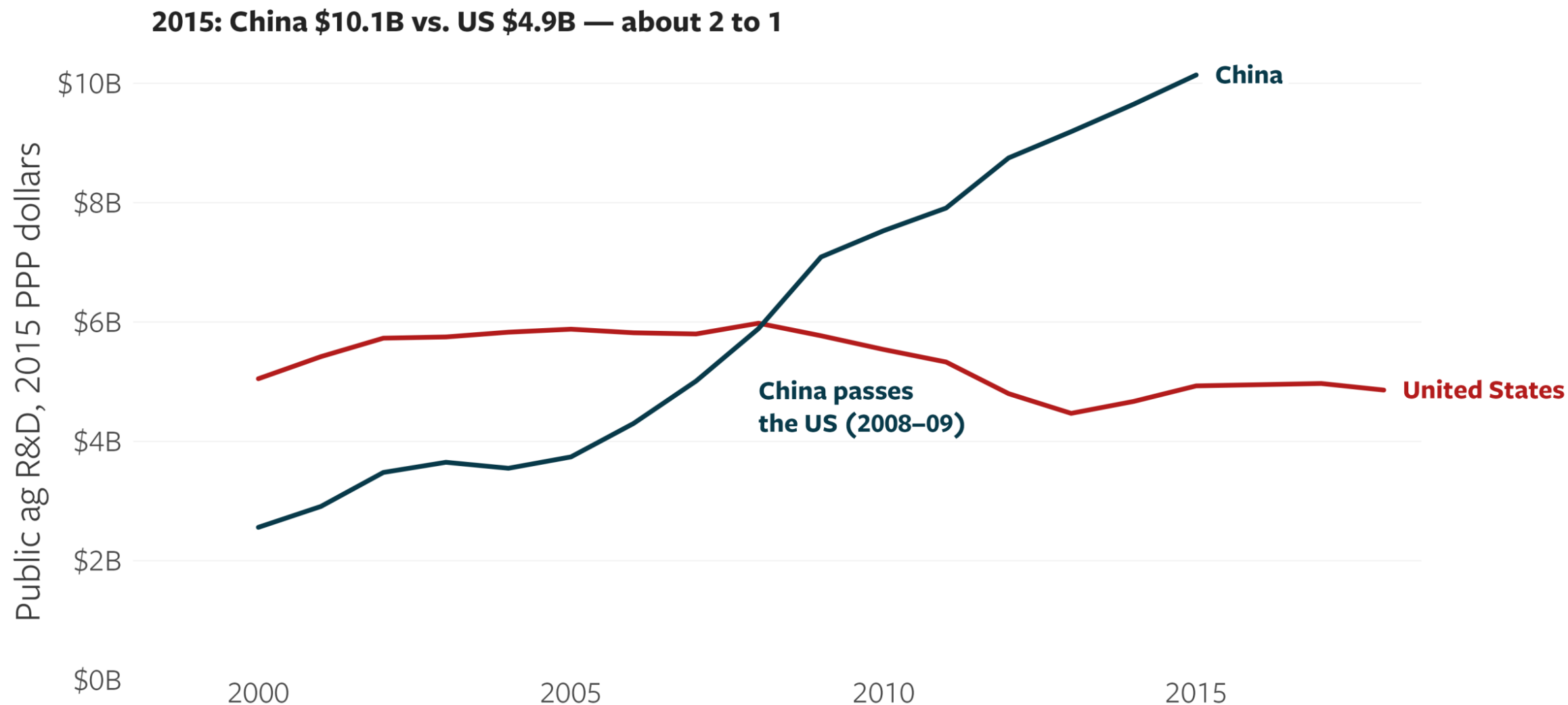
Source: USDA ERS Farm Income and Wealth Statistics (Sept 3, 2026 release; 2026 = forecast); RMA/ERS crop-insurance premium subsidies; ERS public ag R&D.

Public ag R&D: the United States, 2000–2018



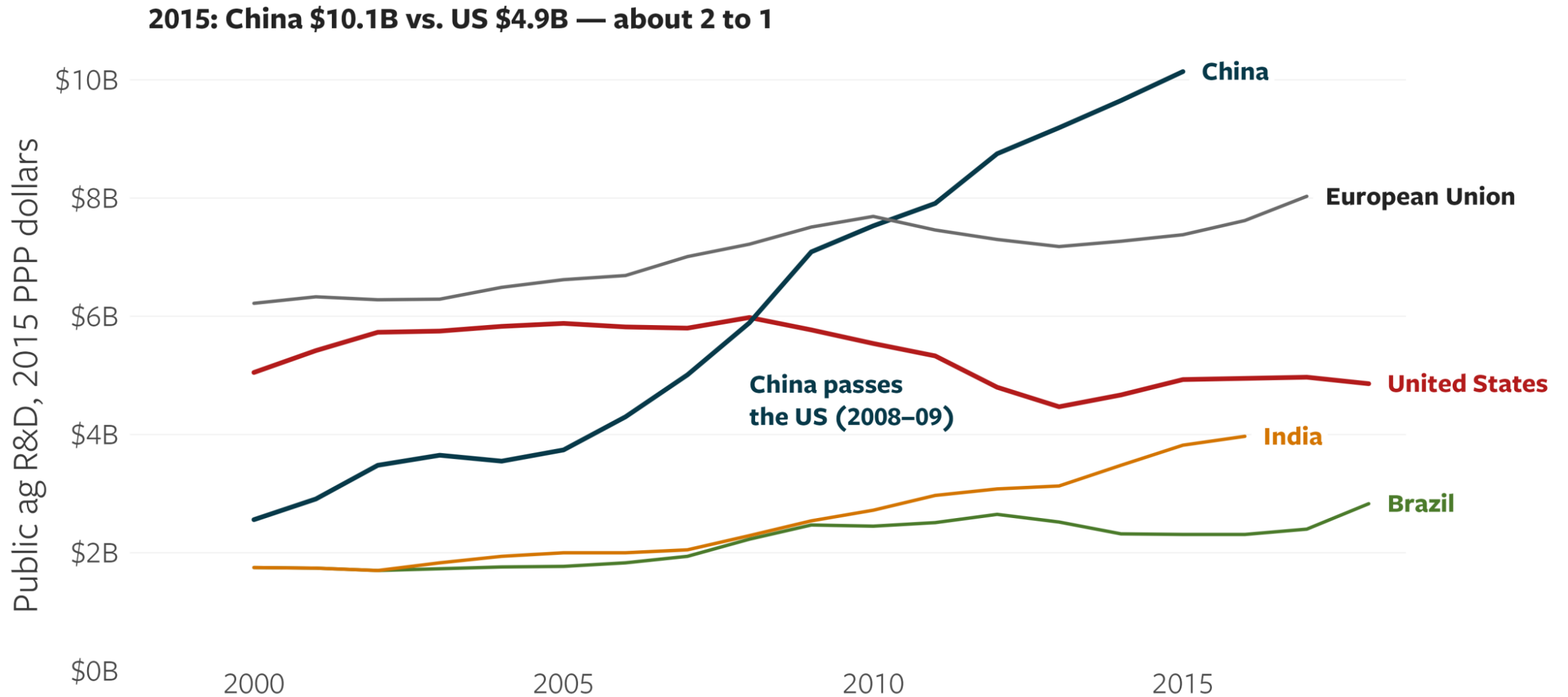
Source: USDA ERS chart of note (2022) based on ERS, ASTI/IFPRI and World Bank data; values digitized from the published chart.

China passed us around 2008; by 2015, two to one



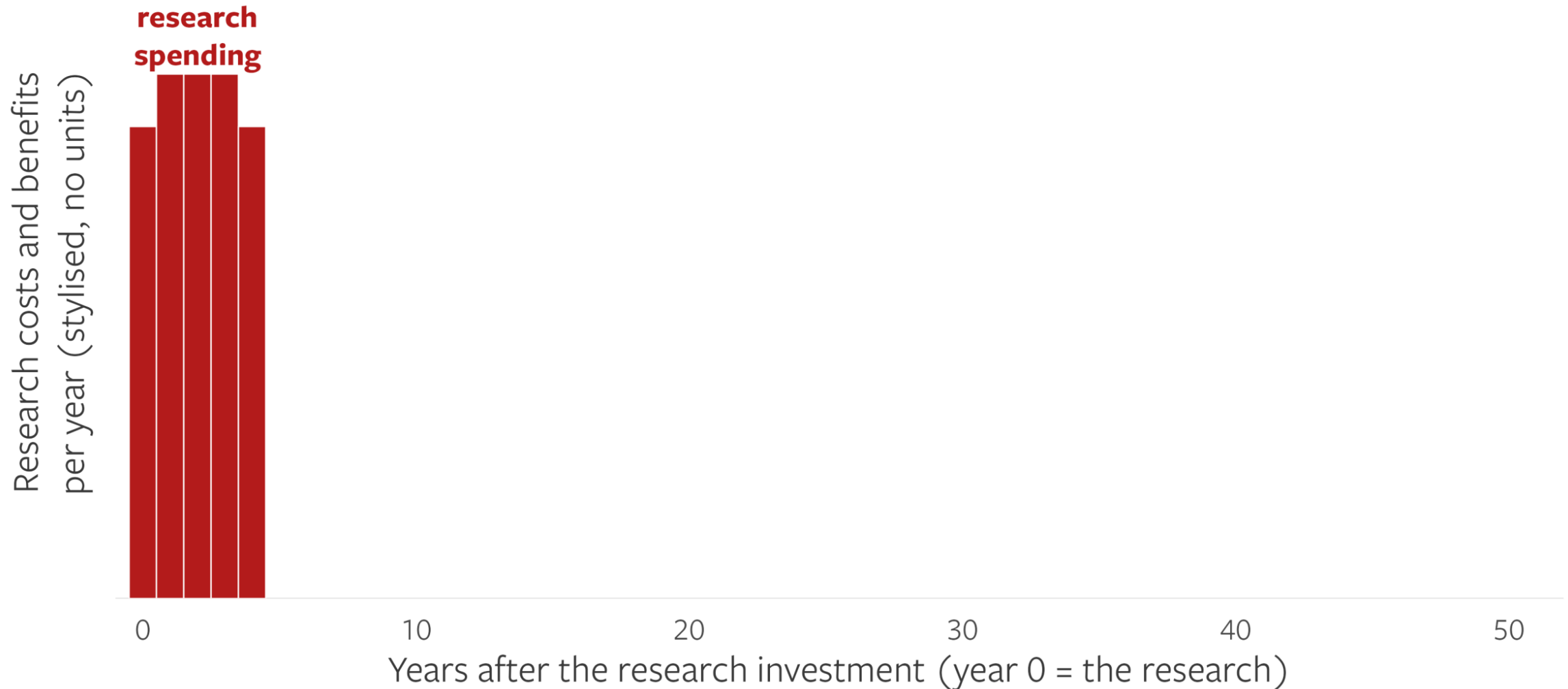
Source: USDA ERS chart of note (2022) based on ERS, ASTI/IFPRI and World Bank data; values digitized from the published chart.

Others are accelerating



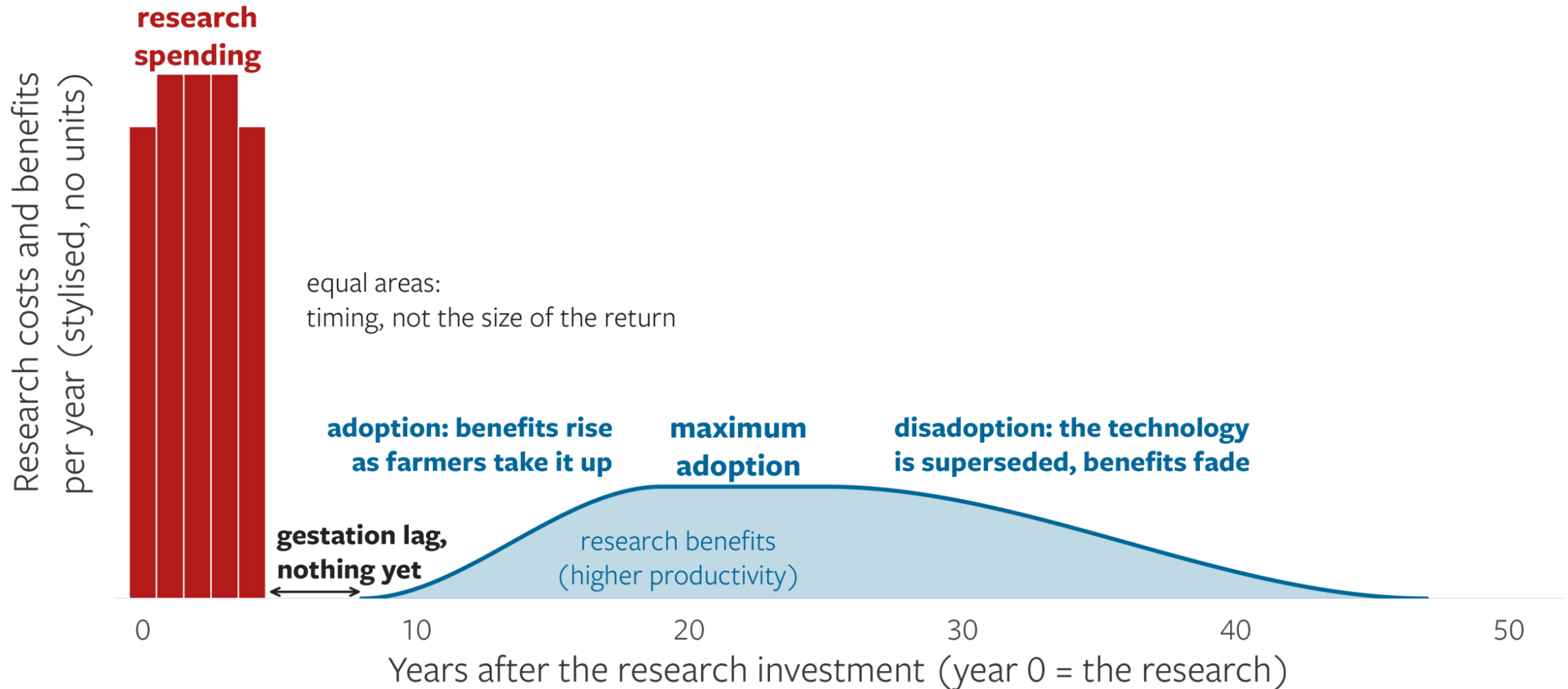
Source: USDA ERS chart of note (2022) based on ERS, ASTI/IFPRI and World Bank data; values digitized from the published chart.

Research is an investment: the money goes in first...



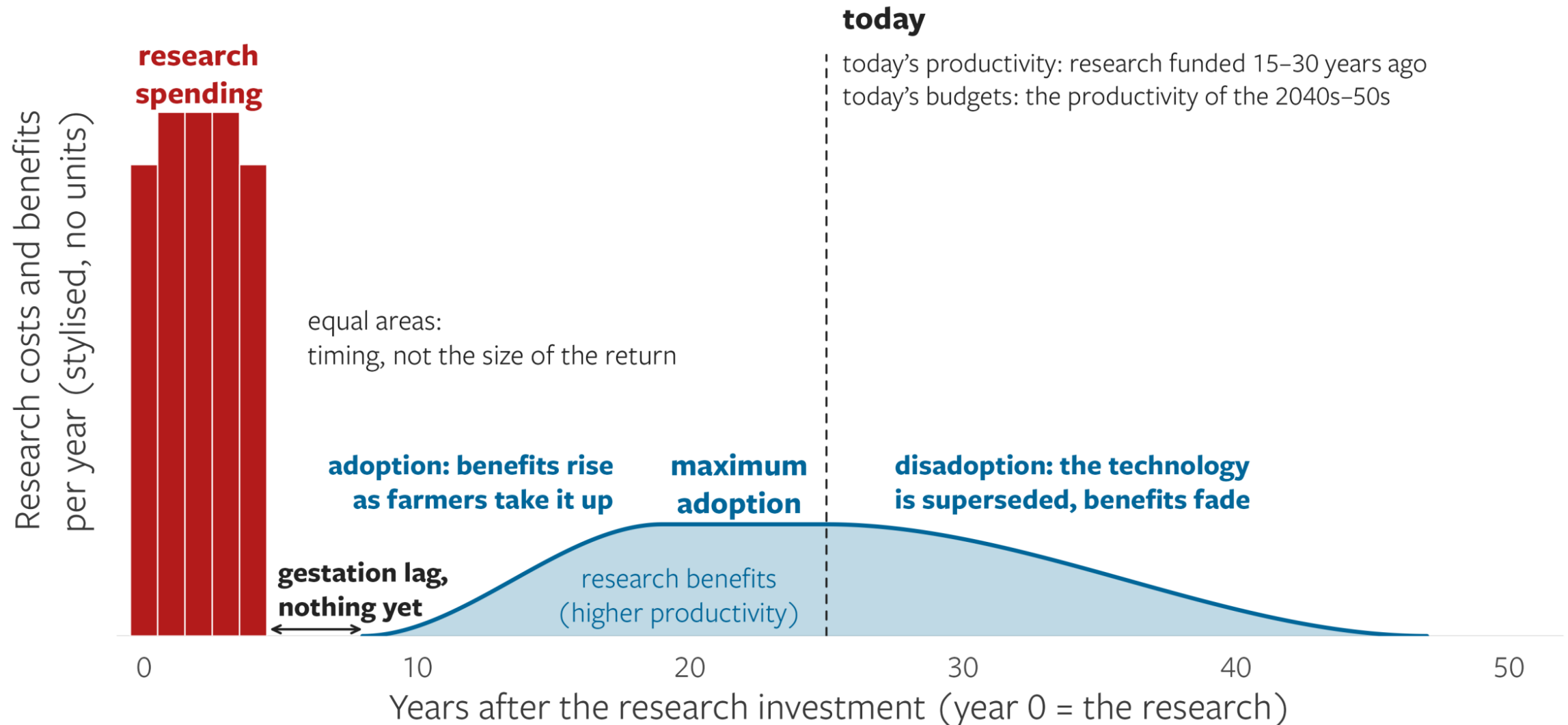
Stylised after Alston, Norton & Pardey (1995) and Alston et al. (2010); shape illustrative, not estimated; the two areas are drawn equal.

...and the benefits arrive over decades



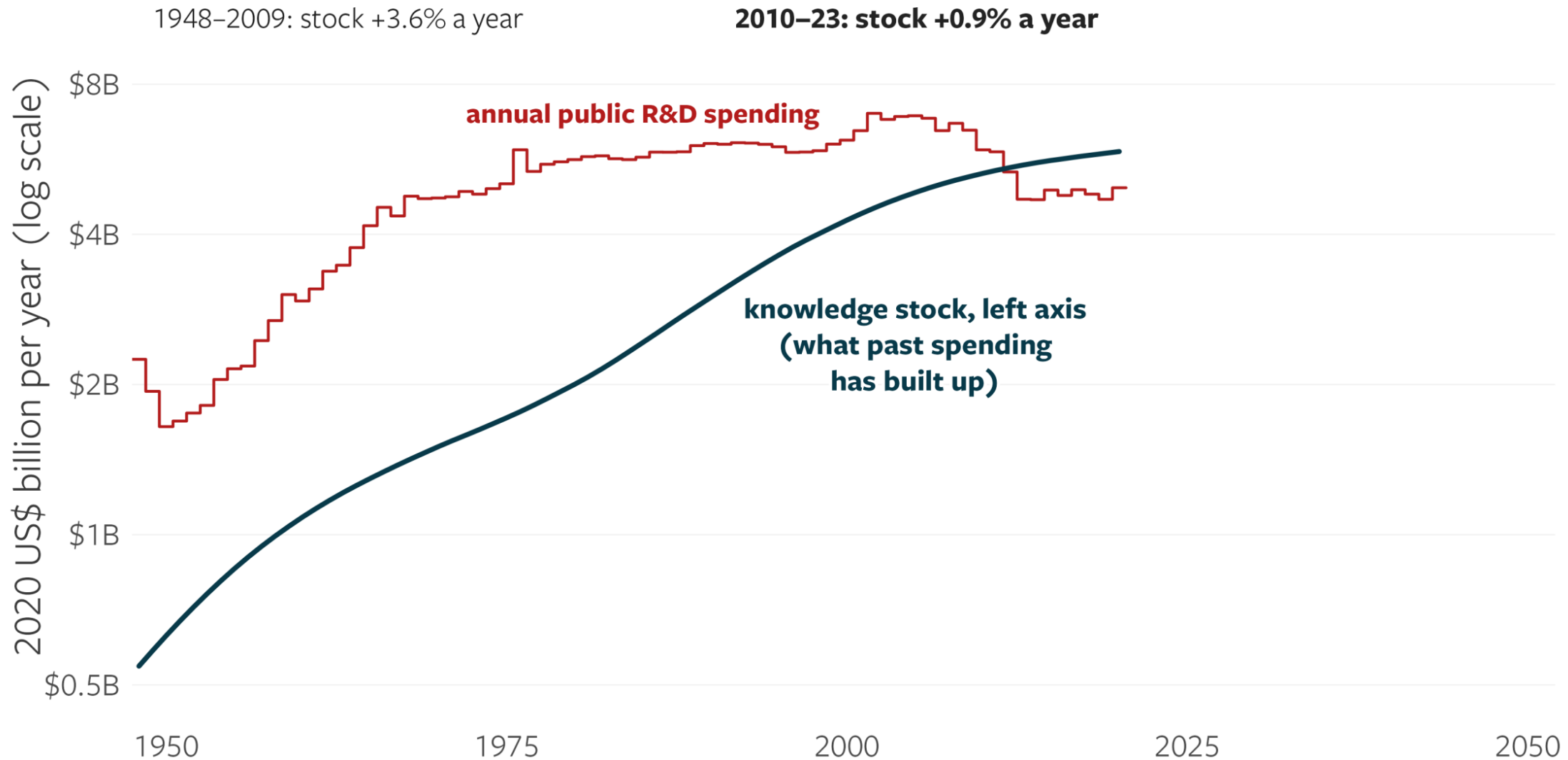
Stylised after Alston, Norton & Pardey (1995) and Alston et al. (2010); shape illustrative, not estimated; the two areas are drawn equal.

Today's yields were bought decades ago



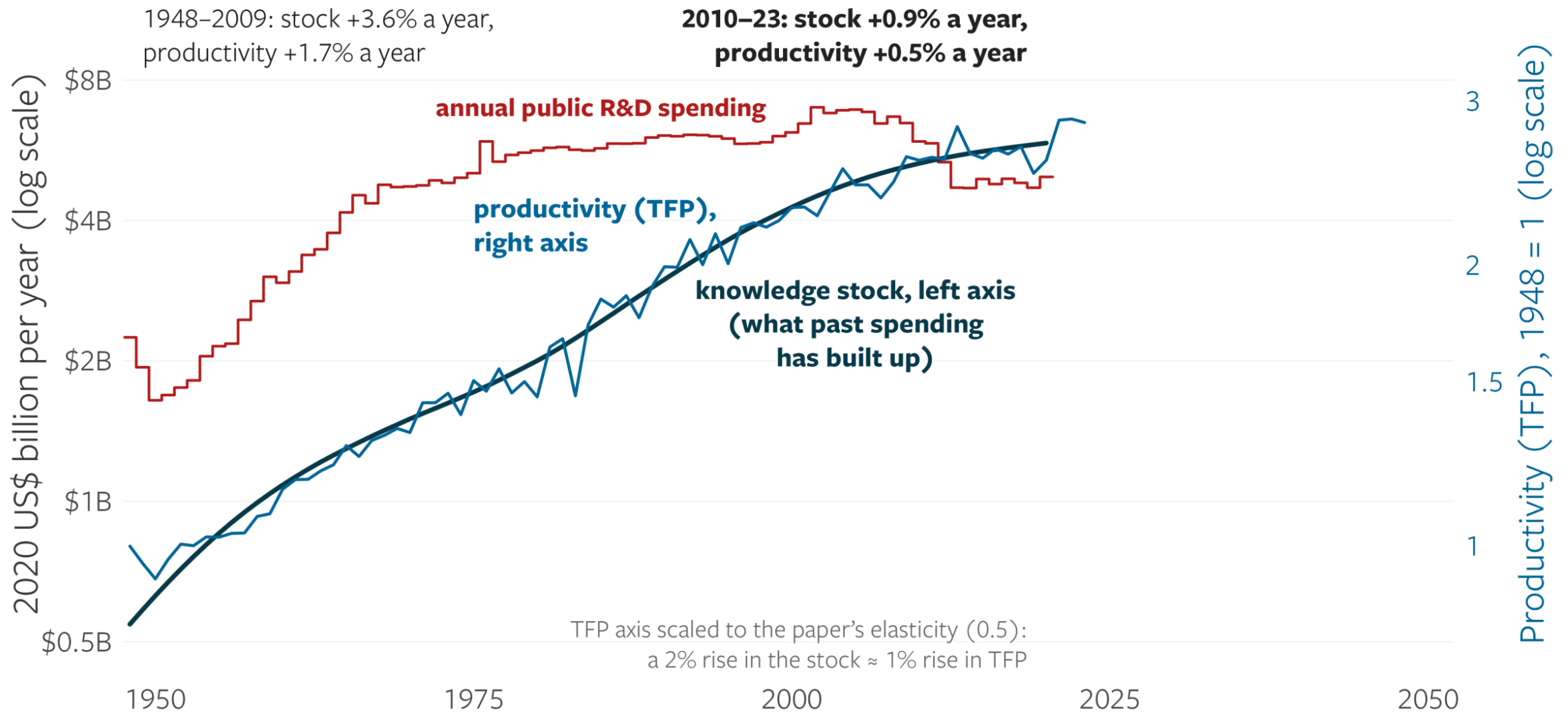
Stylised after Alston, Norton & Pardey (1995) and Alston et al. (2010); shape illustrative, not estimated; the two areas are drawn equal.

Past spending adds up to a knowledge stock — which has stopped growing



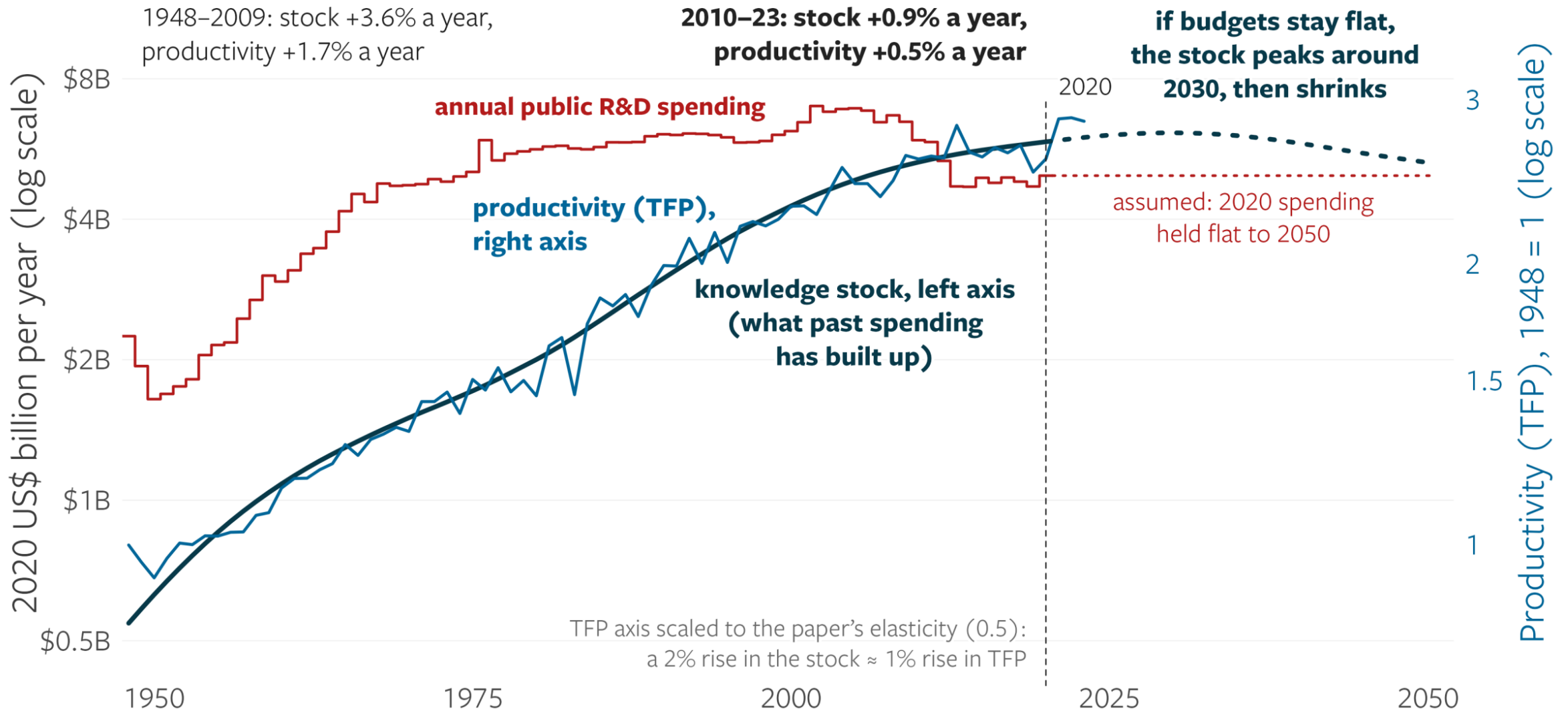
Source: USDA ERS spending and TFP; stock = 50-year weighted average of past spending with the baseline lag weights; TFP axis at elasticity 0.5 (Ortiz-Bobea et al. 2025).

Productivity tracks the knowledge stock



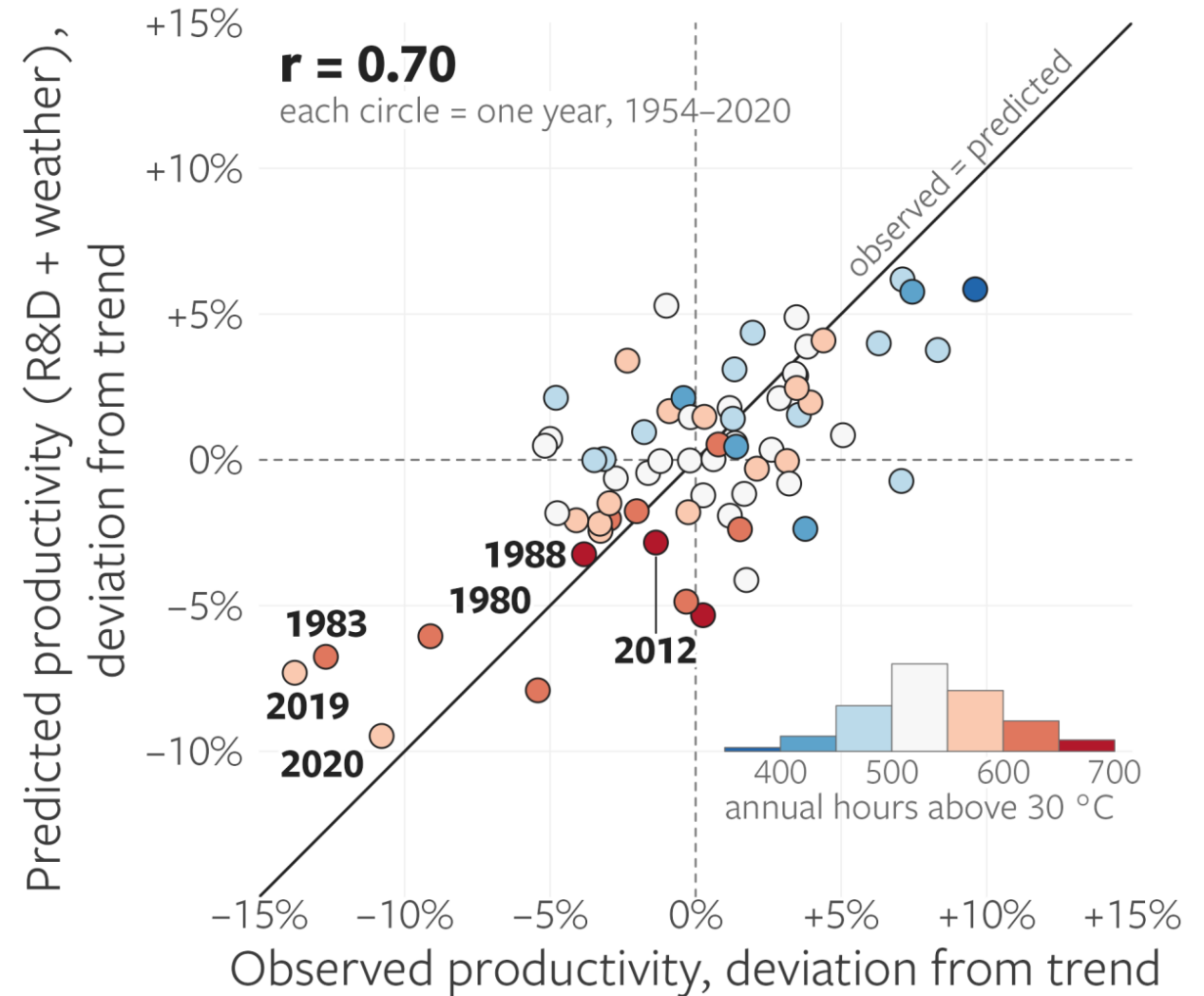
Source: USDA ERS spending and TFP; stock = 50-year weighted average of past spending with the baseline lag weights; TFP axis at elasticity 0.5 (Ortiz-Bobea et al. 2025).

Flat budgets: the stock peaks around 2030, then shrinks

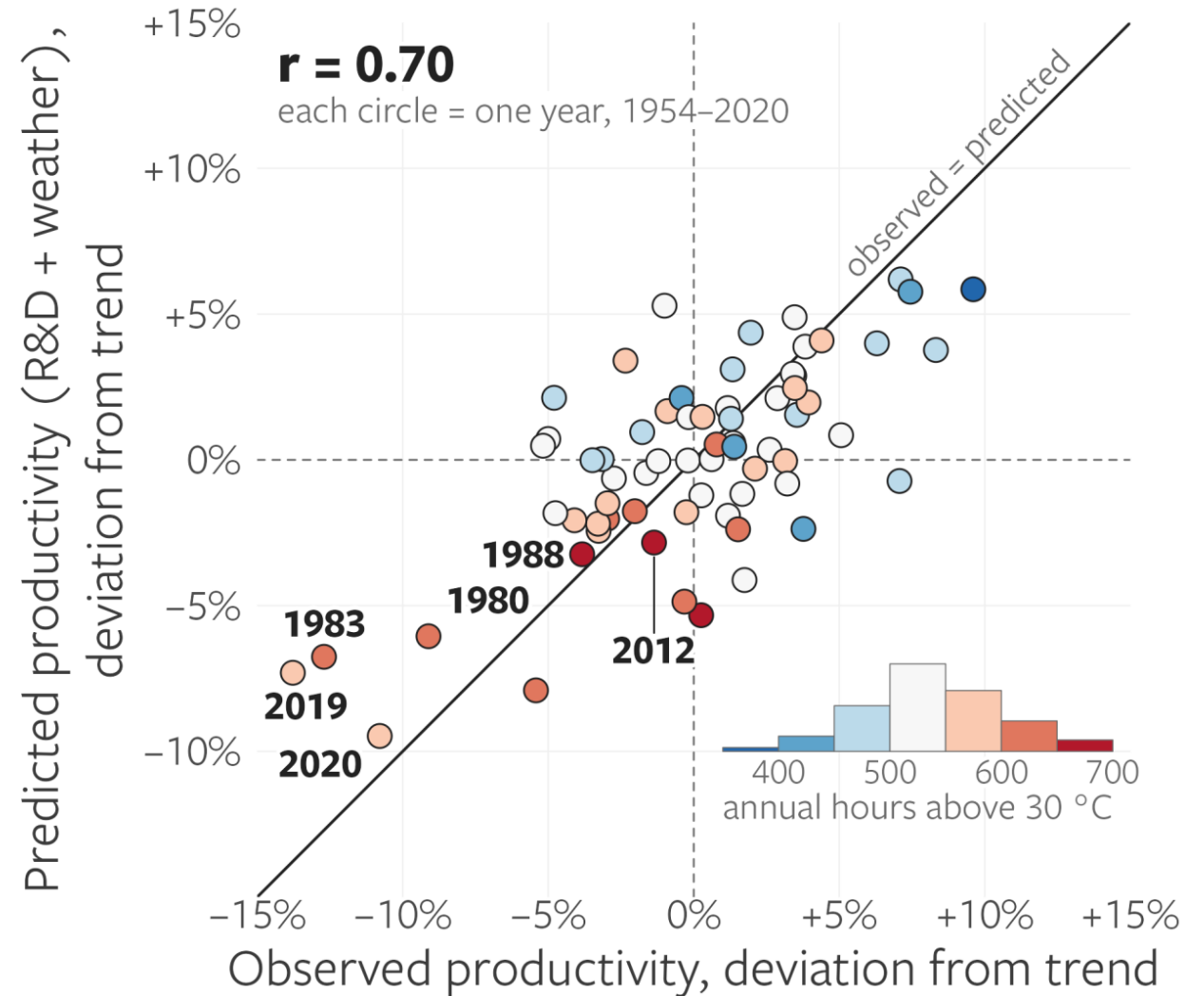
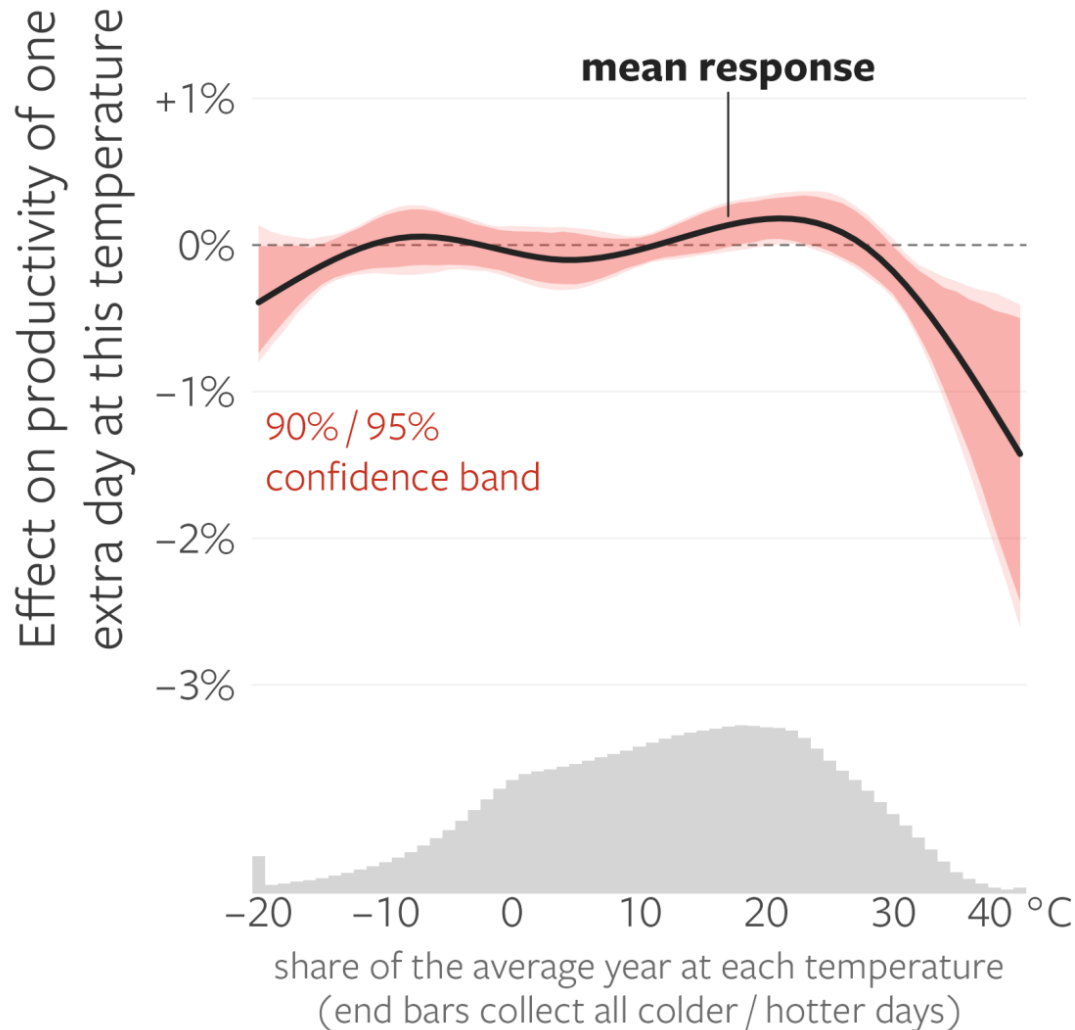


Source: USDA ERS spending and TFP; stock = 50-year weighted average of past spending with the baseline lag weights; TFP axis at elasticity 0.5 (Ortiz-Bobea et al. 2025).

Year to year, productivity swings with the weather — and the bad years are the hot ones

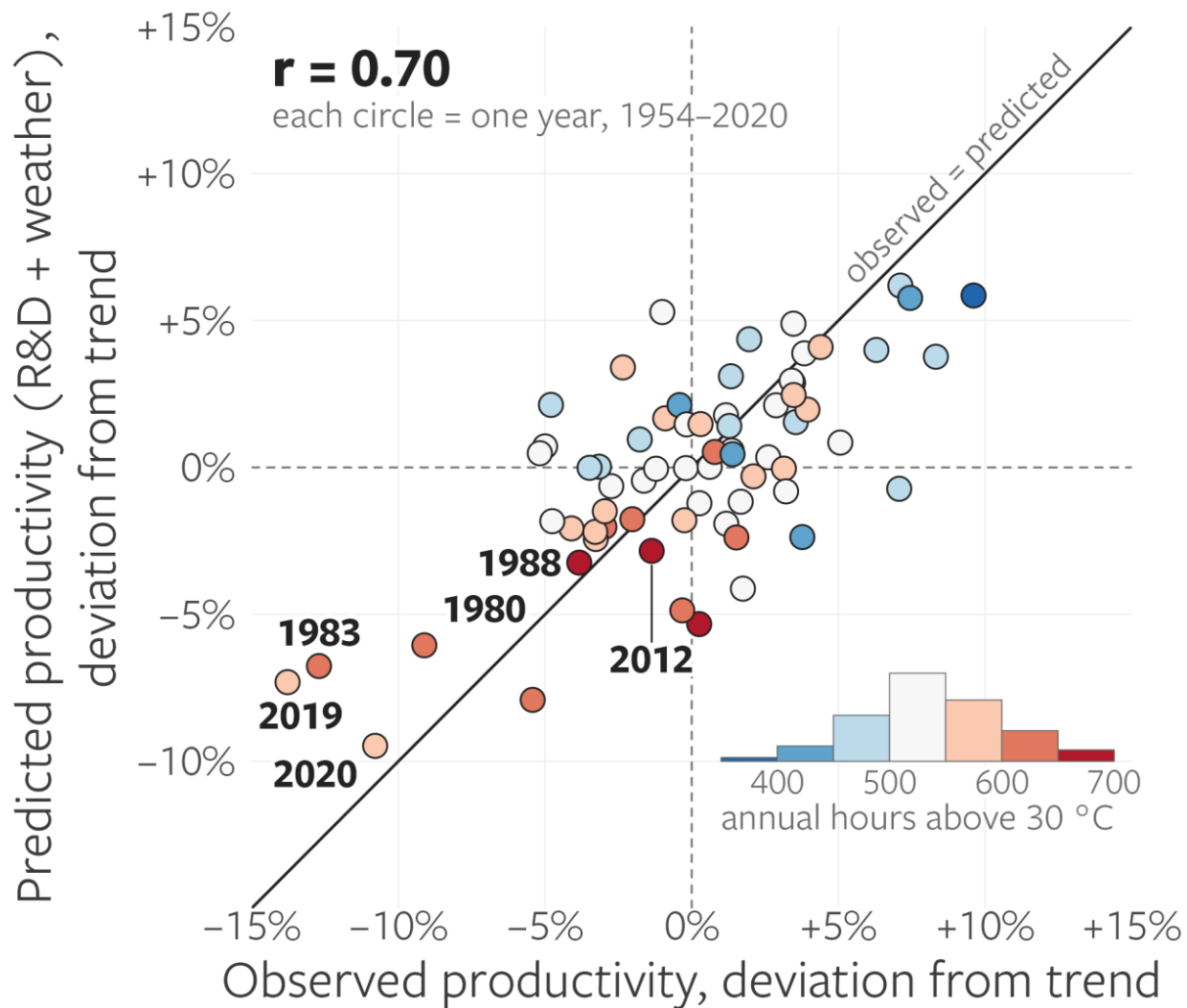
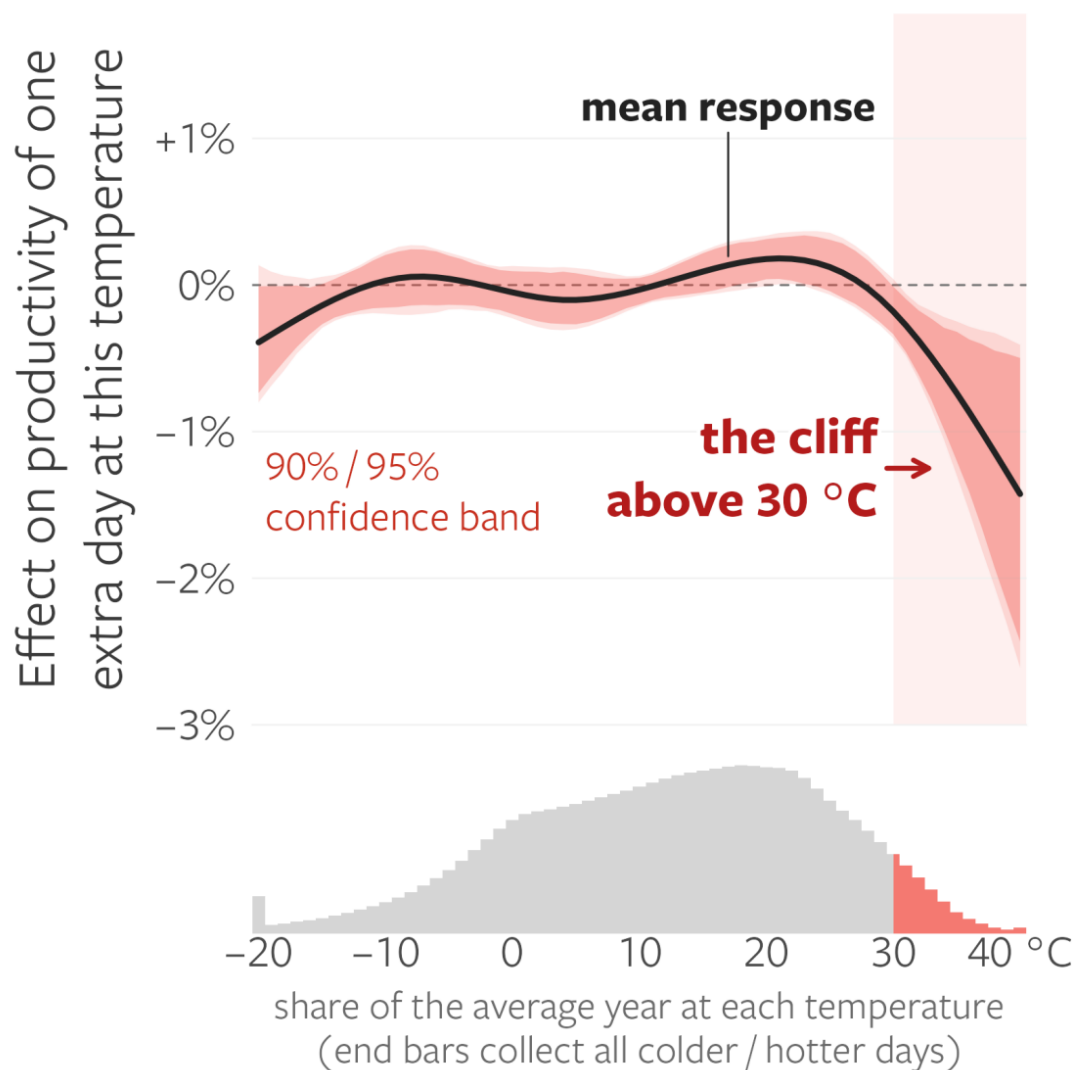


Increasingly, it is climate: heat hurts productivity



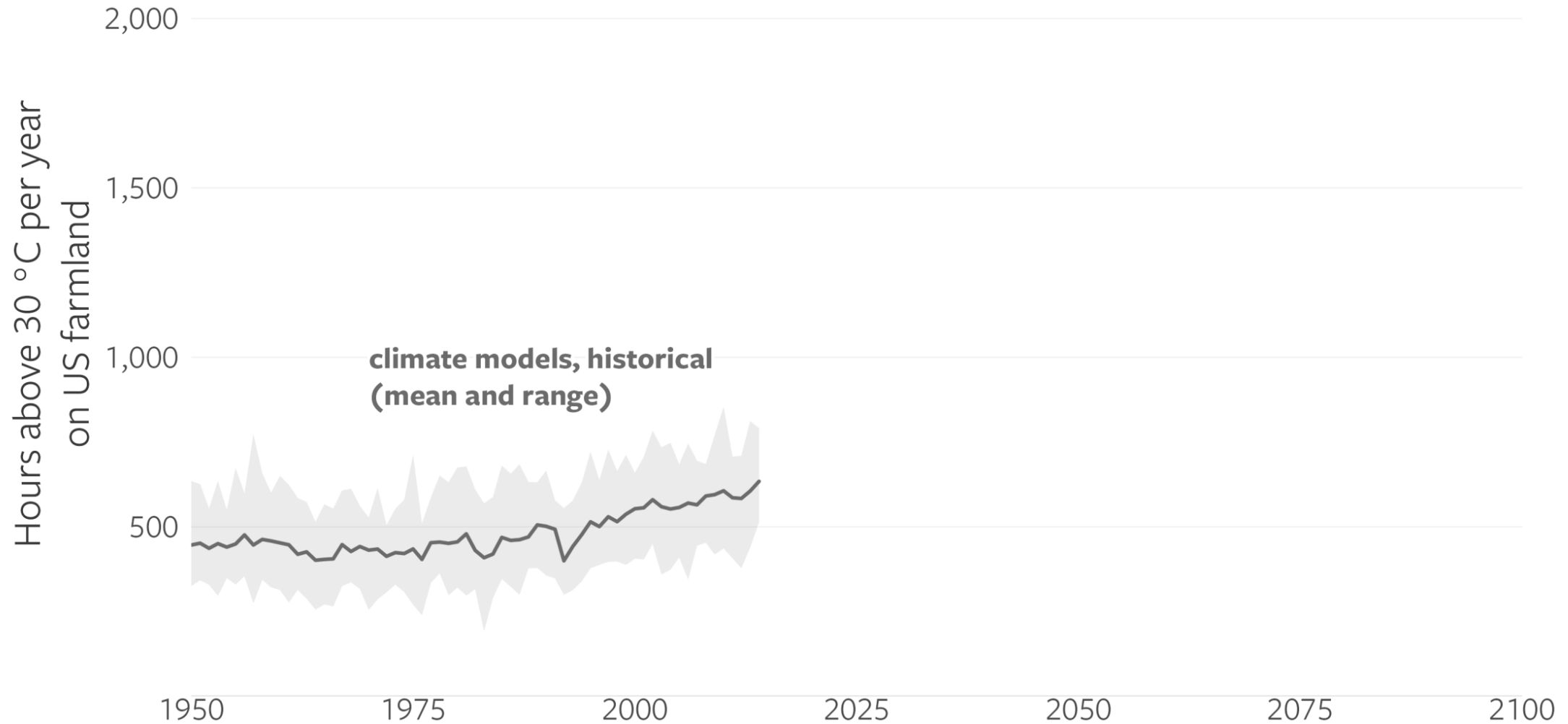
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 2A–B.

Above 30 °C, productivity falls off a cliff



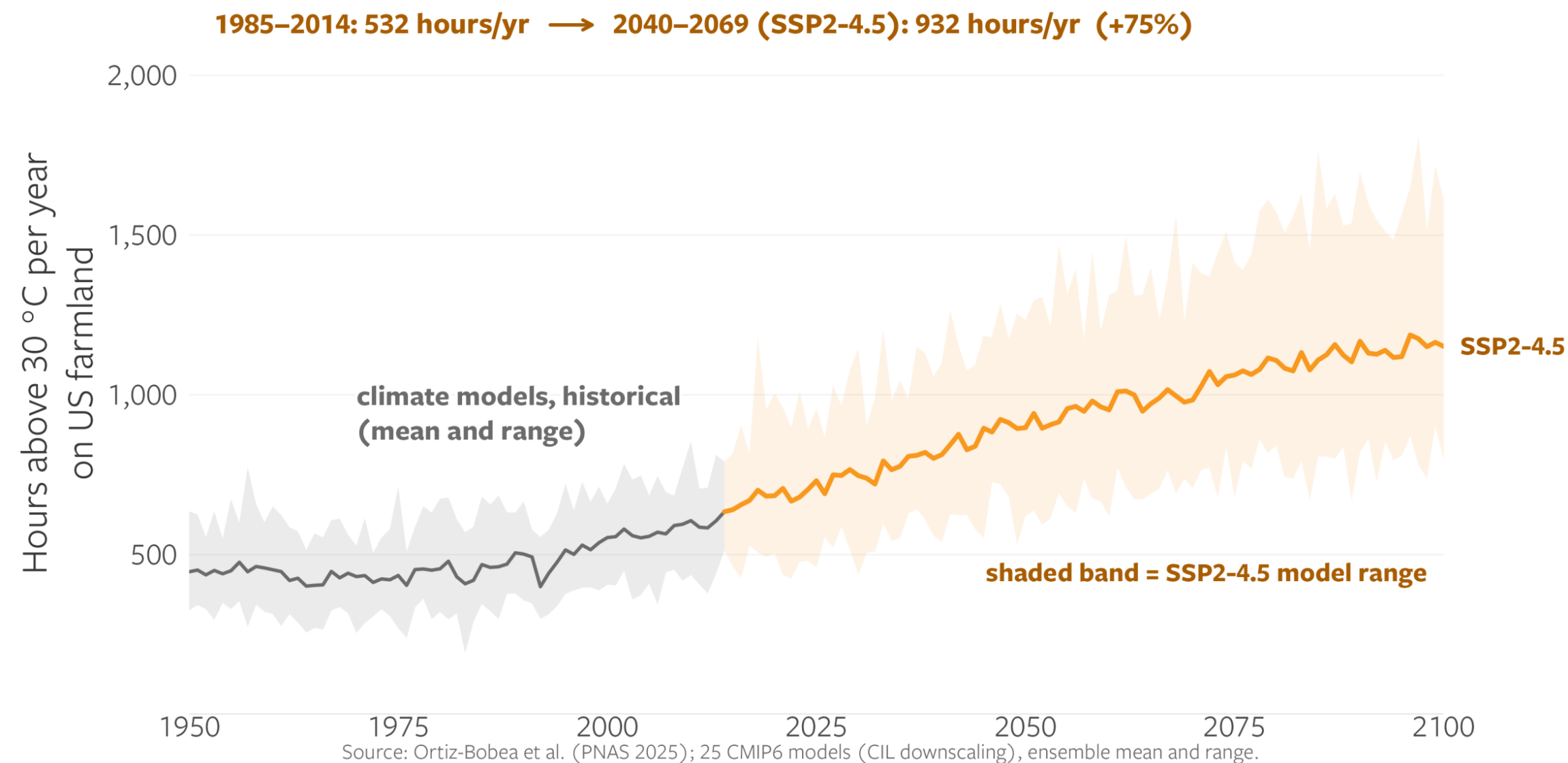
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 2A-B.

Hours above 30 °C over US farmland: the climate models' history

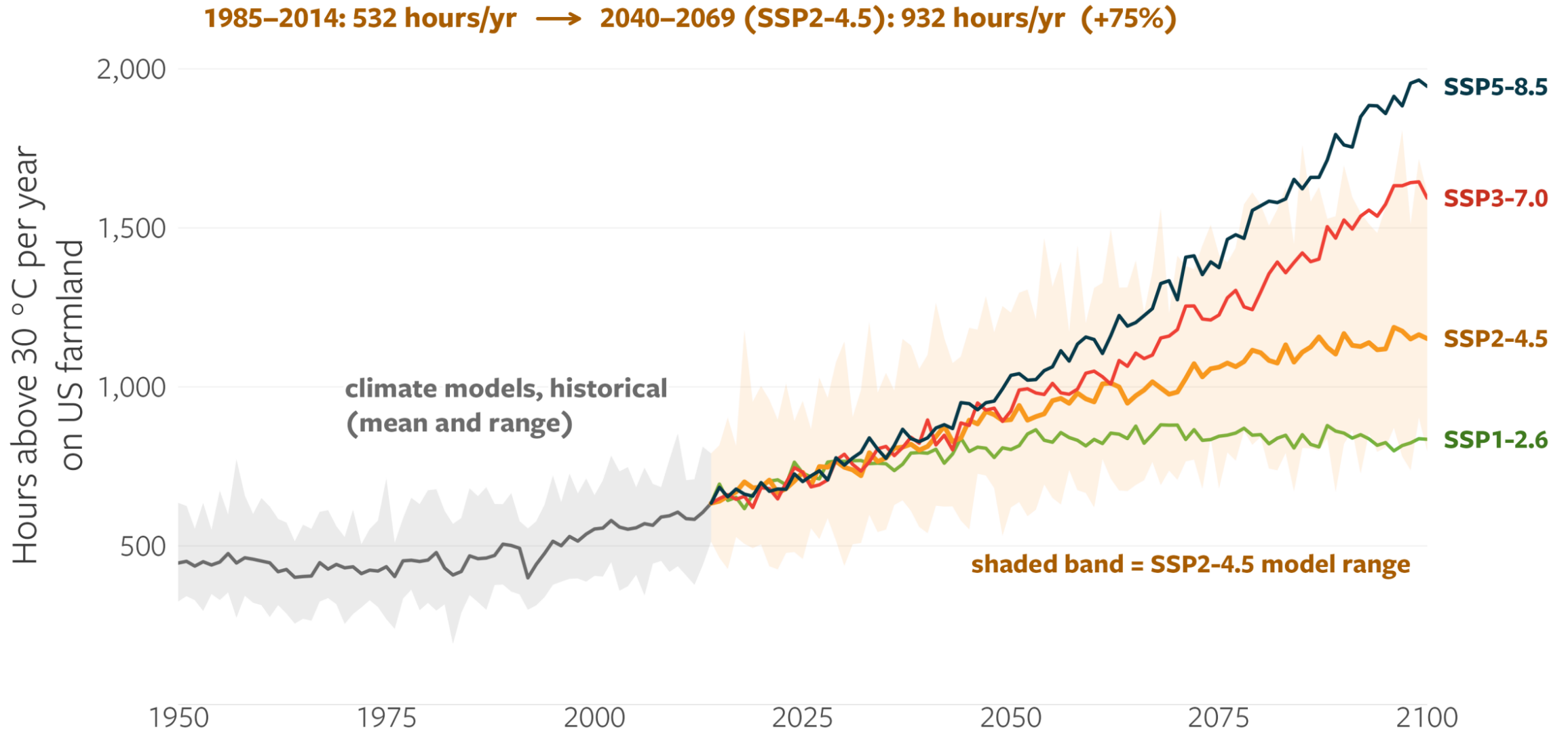


Source: Ortiz-Bobea et al. (PNAS 2025); 25 CMIP6 models (CIL downscaling), ensemble mean and range.

Climate is pushing crops toward the cliff

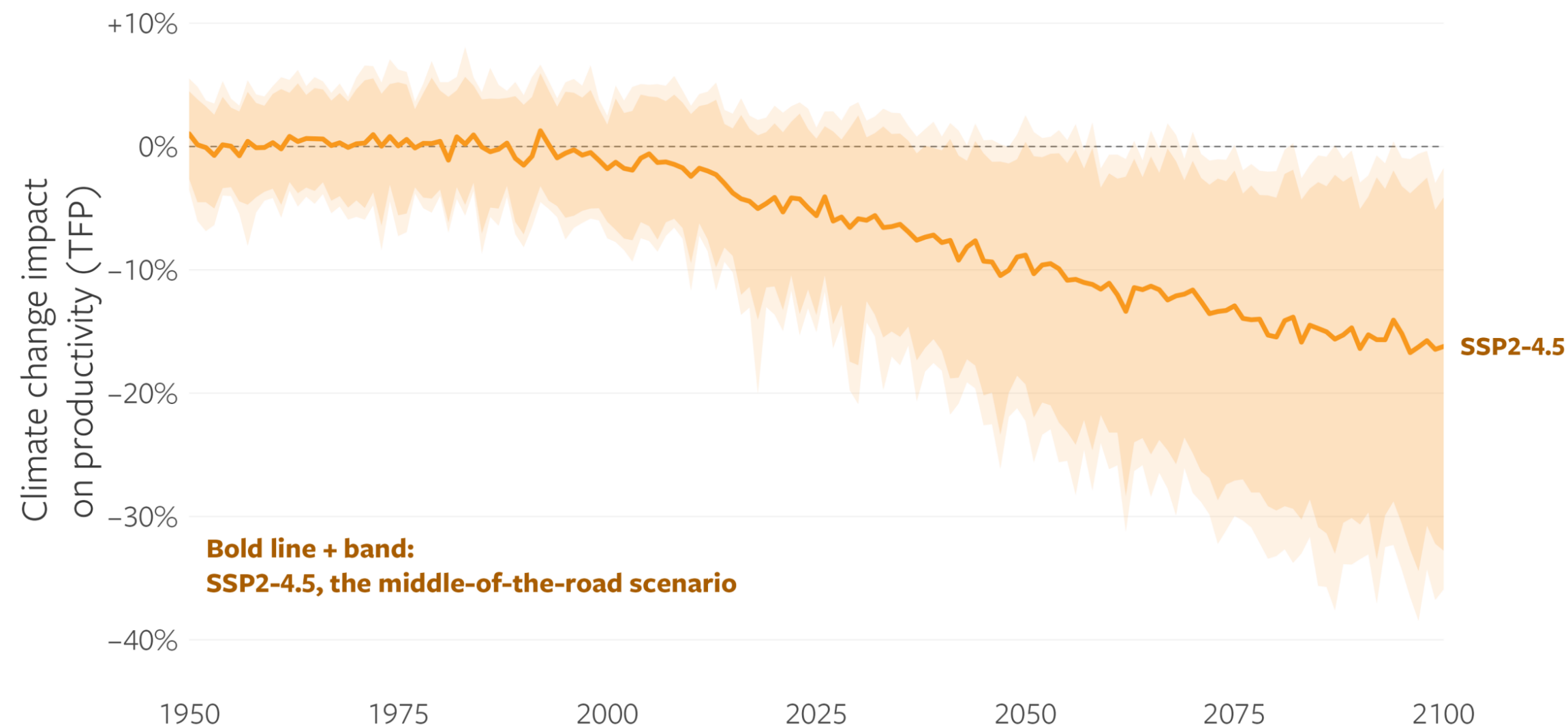


Under every scenario



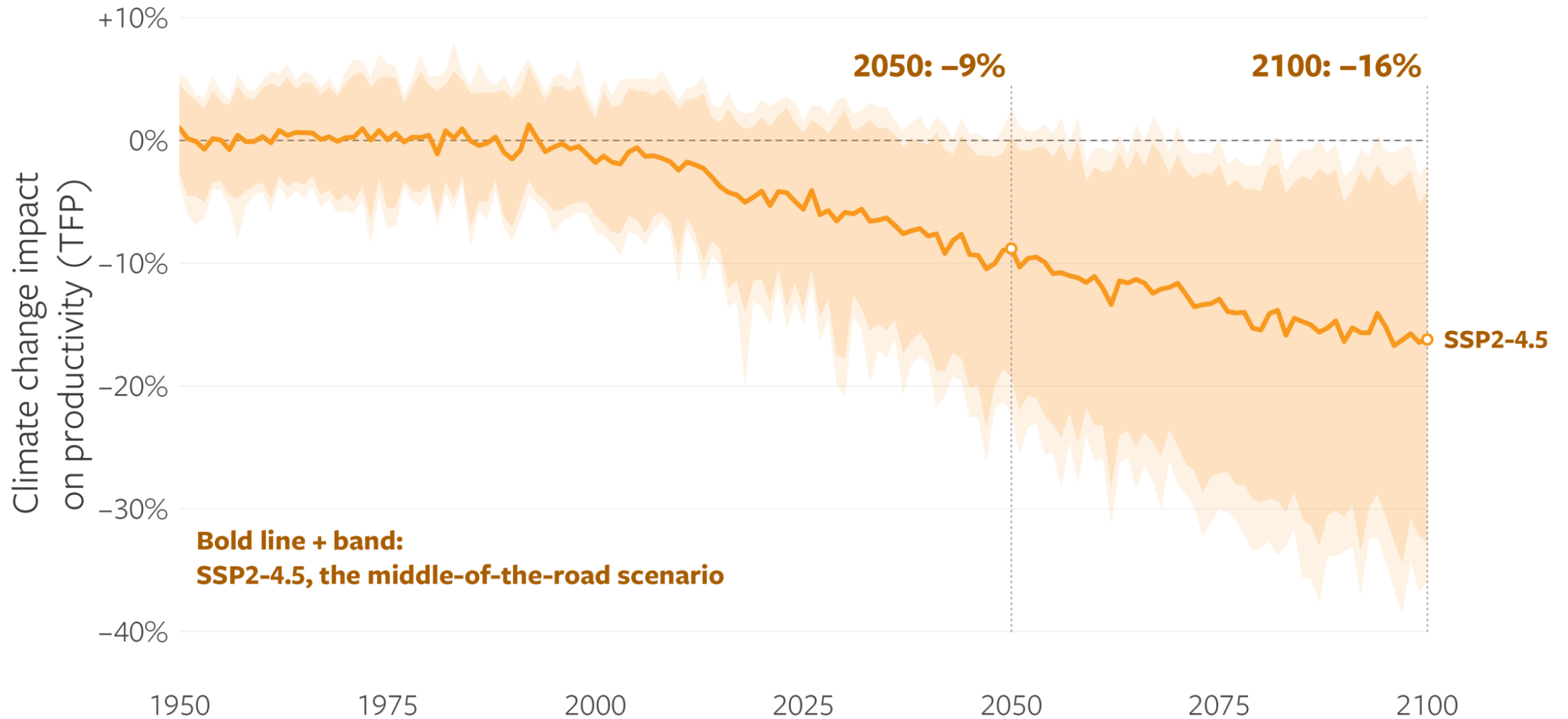
Source: Ortiz-Bobea et al. (PNAS 2025); 25 CMIP6 models (CIL downscaling), ensemble mean and range.

What that does to productivity



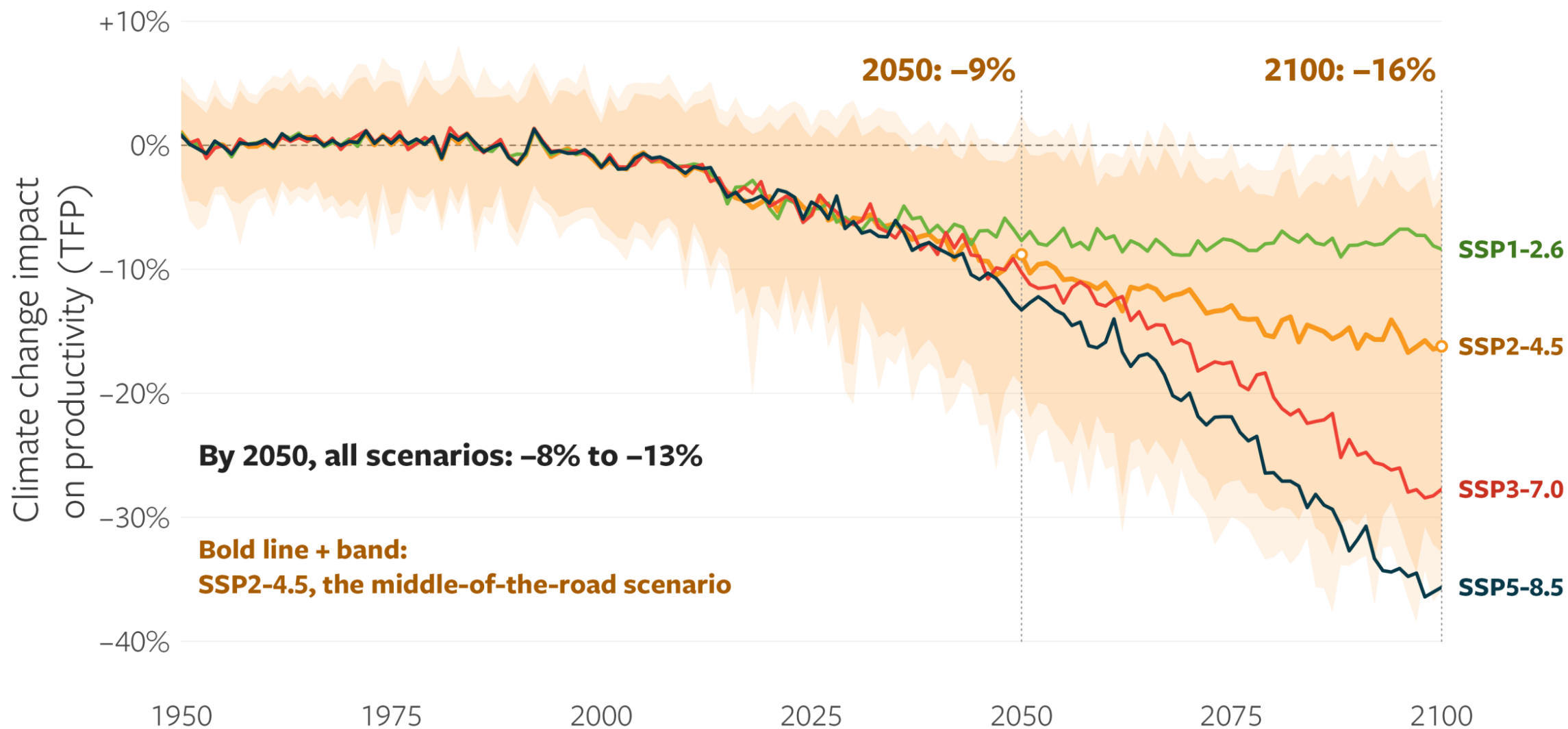
Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 3: CMIP6 ensemble with bootstrap uncertainty (90% and 95% bands).

About 9% lower by 2050



Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 3: CMIP6 ensemble with bootstrap uncertainty (90% and 95% bands).

8–13% lower by 2050 — under every scenario



Source: Ortiz-Bobea et al. (PNAS 2025), Fig. 3: CMIP6 ensemble with bootstrap uncertainty (90% and 95% bands).

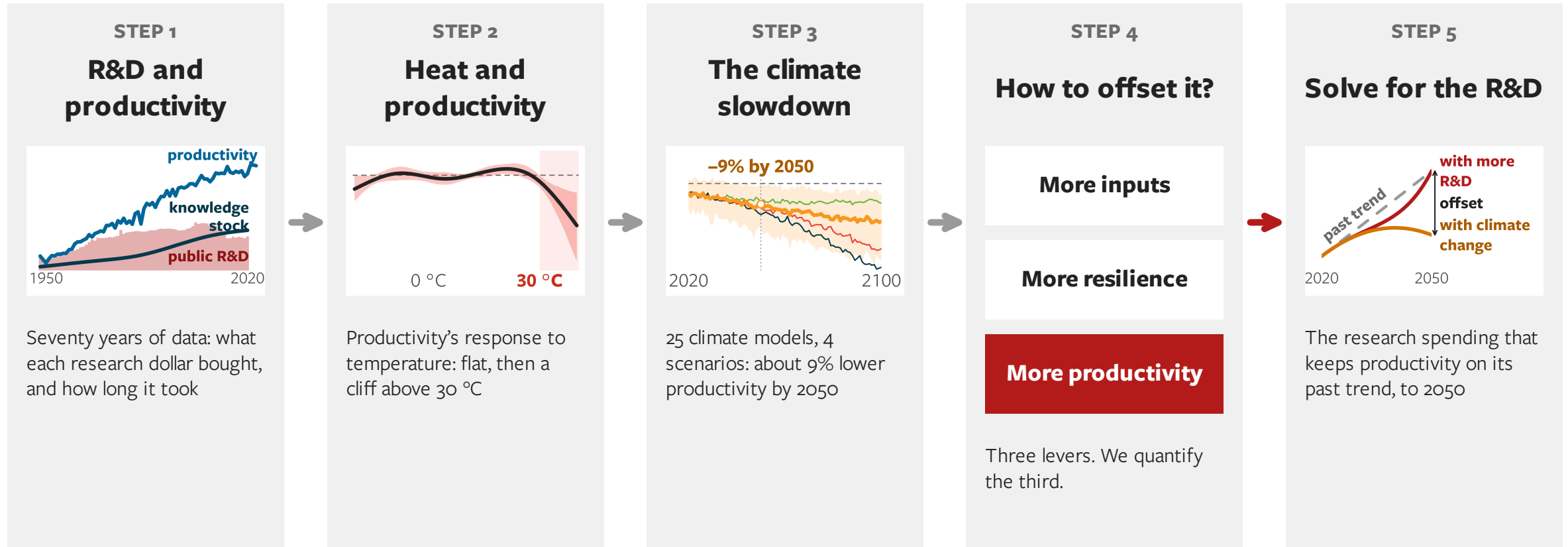
Message 2 — recap

- 1 Productivity growth is slowing.
- 2 **The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.

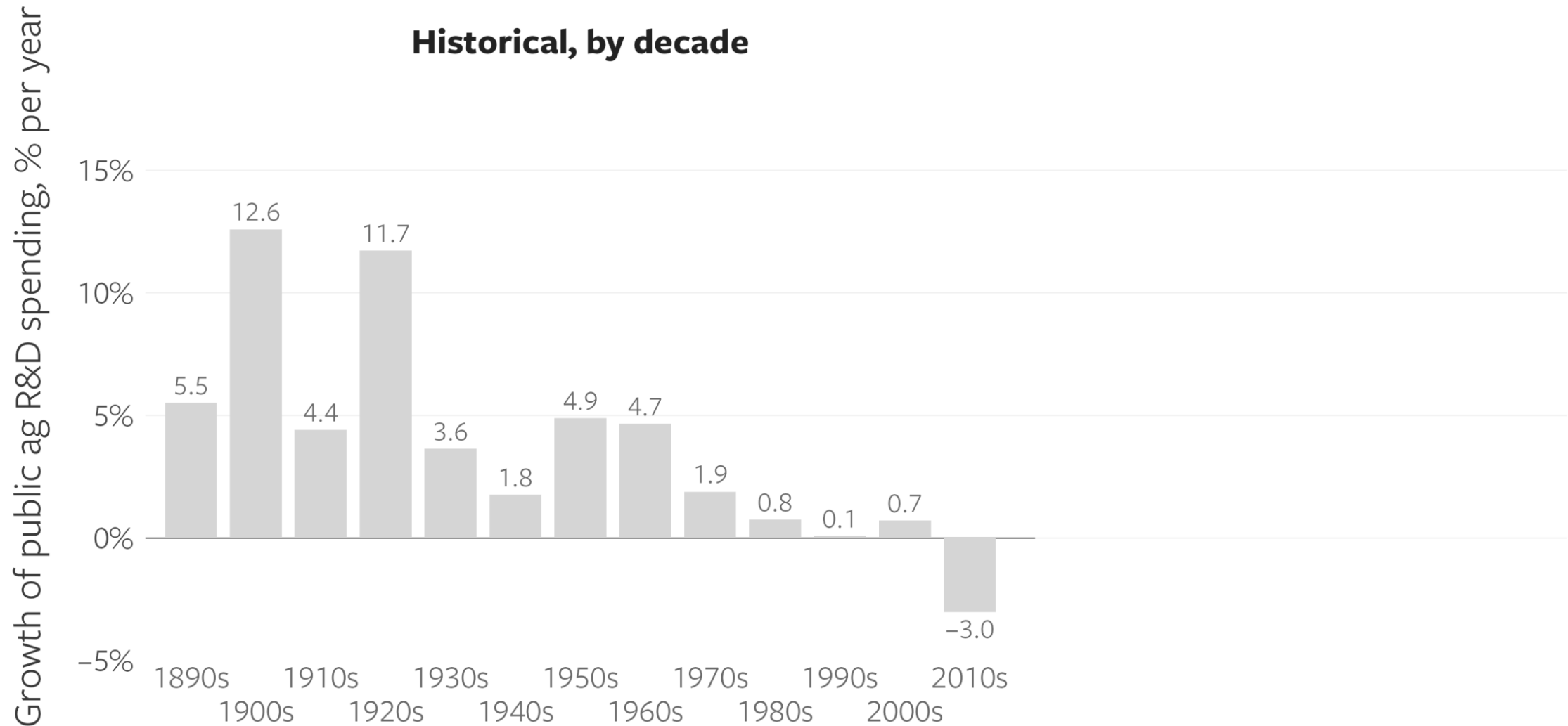
Message 3

- 1 Productivity growth is slowing.
- 2 The main driver is slowing R&D — but increasingly it's climate.
- 3 **Maintaining productivity growth requires a historical pivot.**

What we did in the PNAS 2025 paper, in five steps

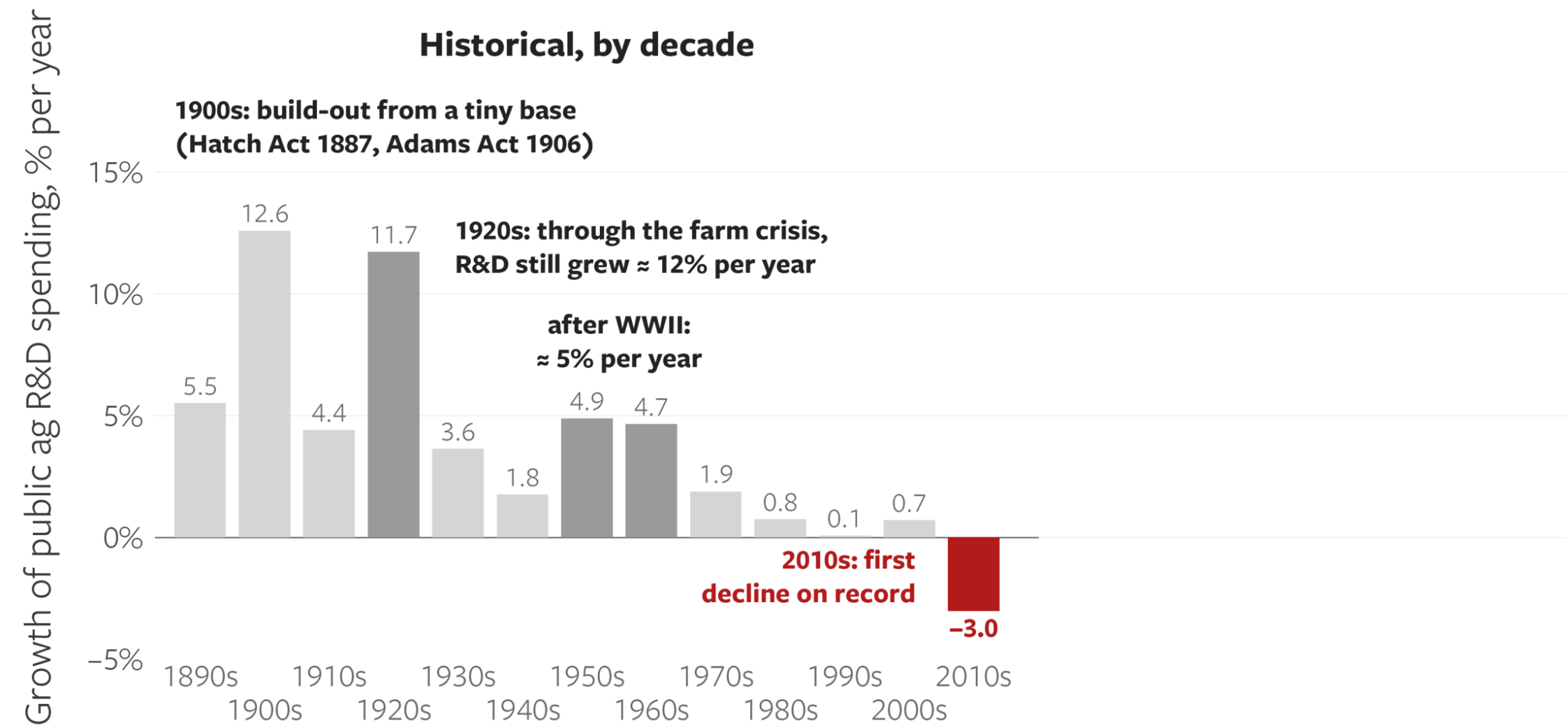


Strategy 1: gradual growth. Has it been done before?



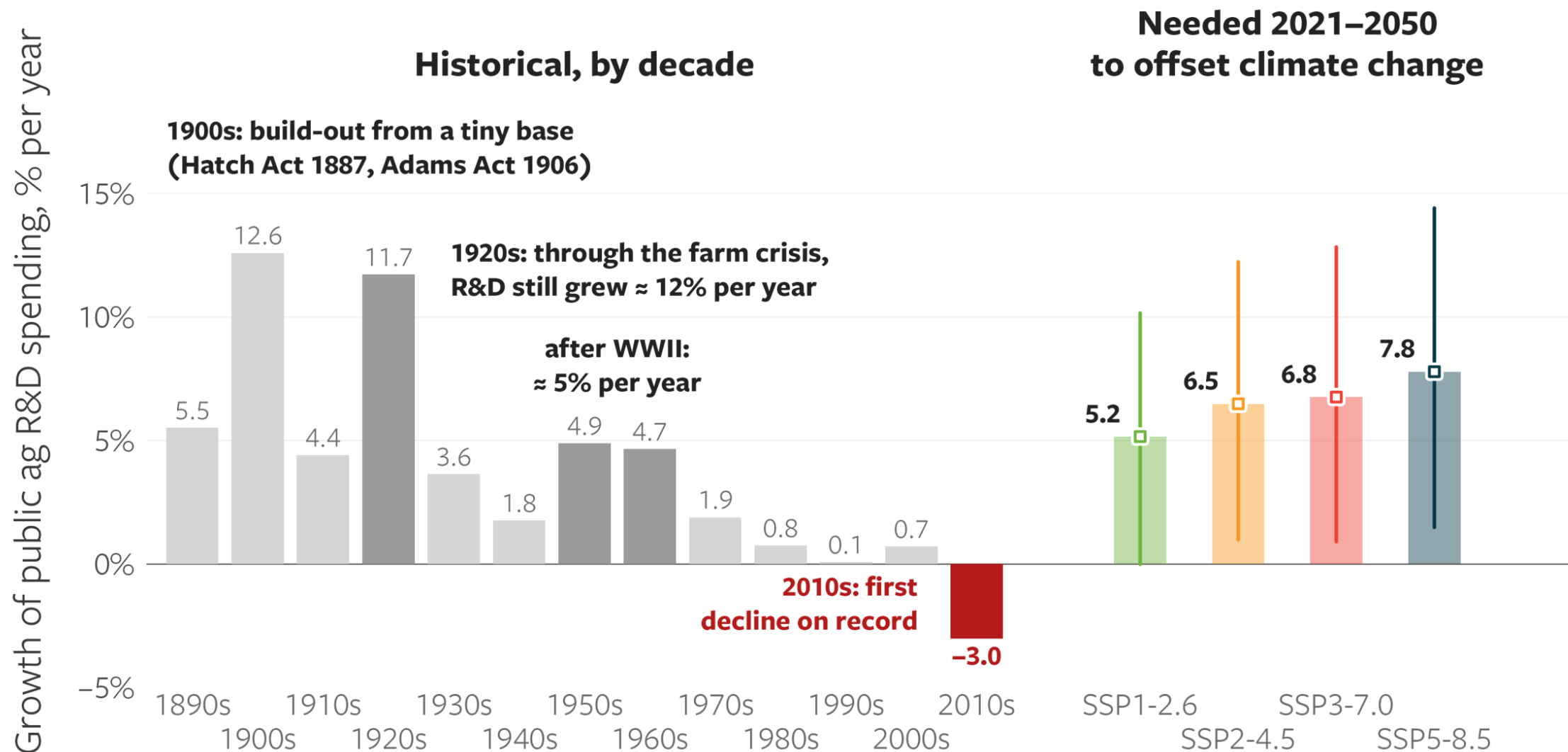
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4B and Table S12 (means and 95% CI).

We have done this before. Twice.



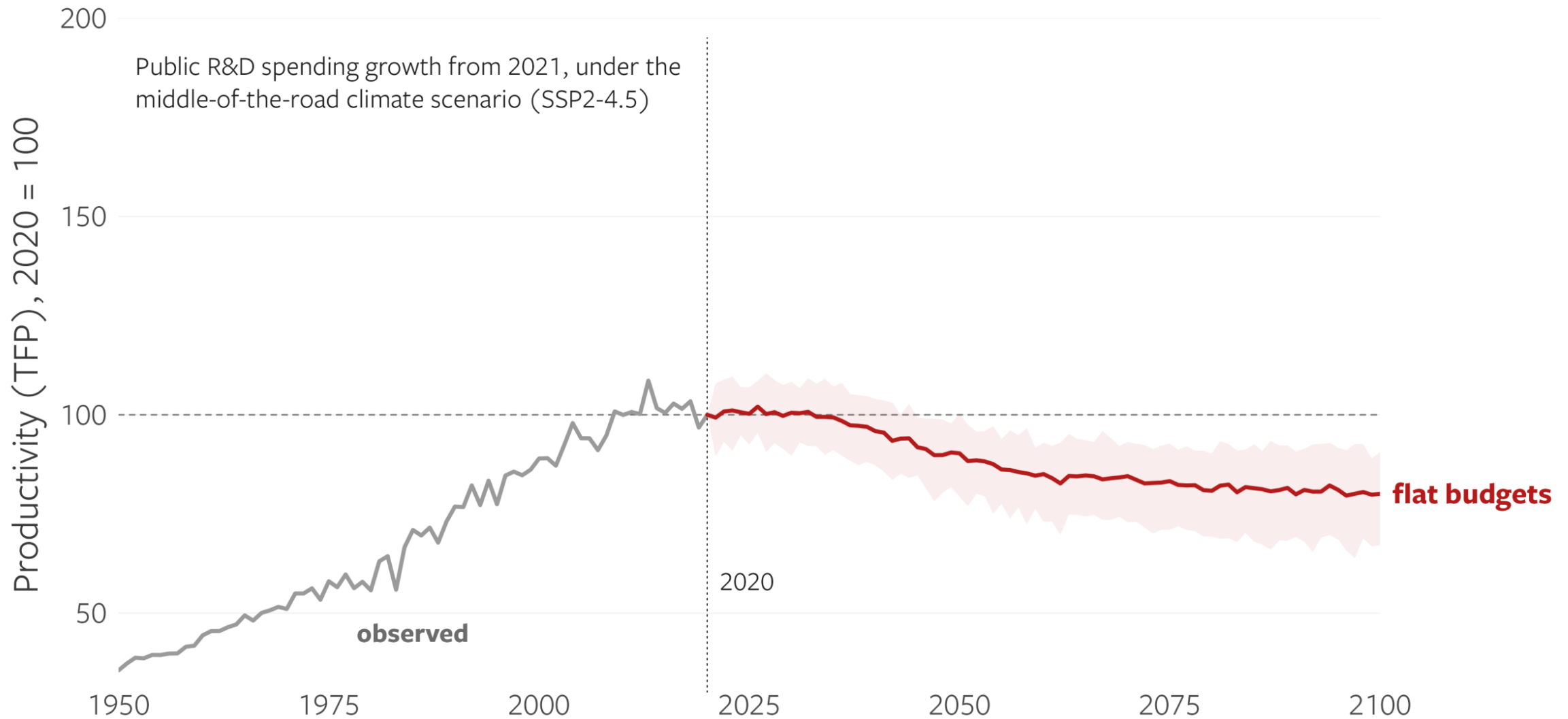
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4B and Table S12 (means and 95% CI).

To offset climate by 2050: 5–8% a year, a 1920s-style pivot



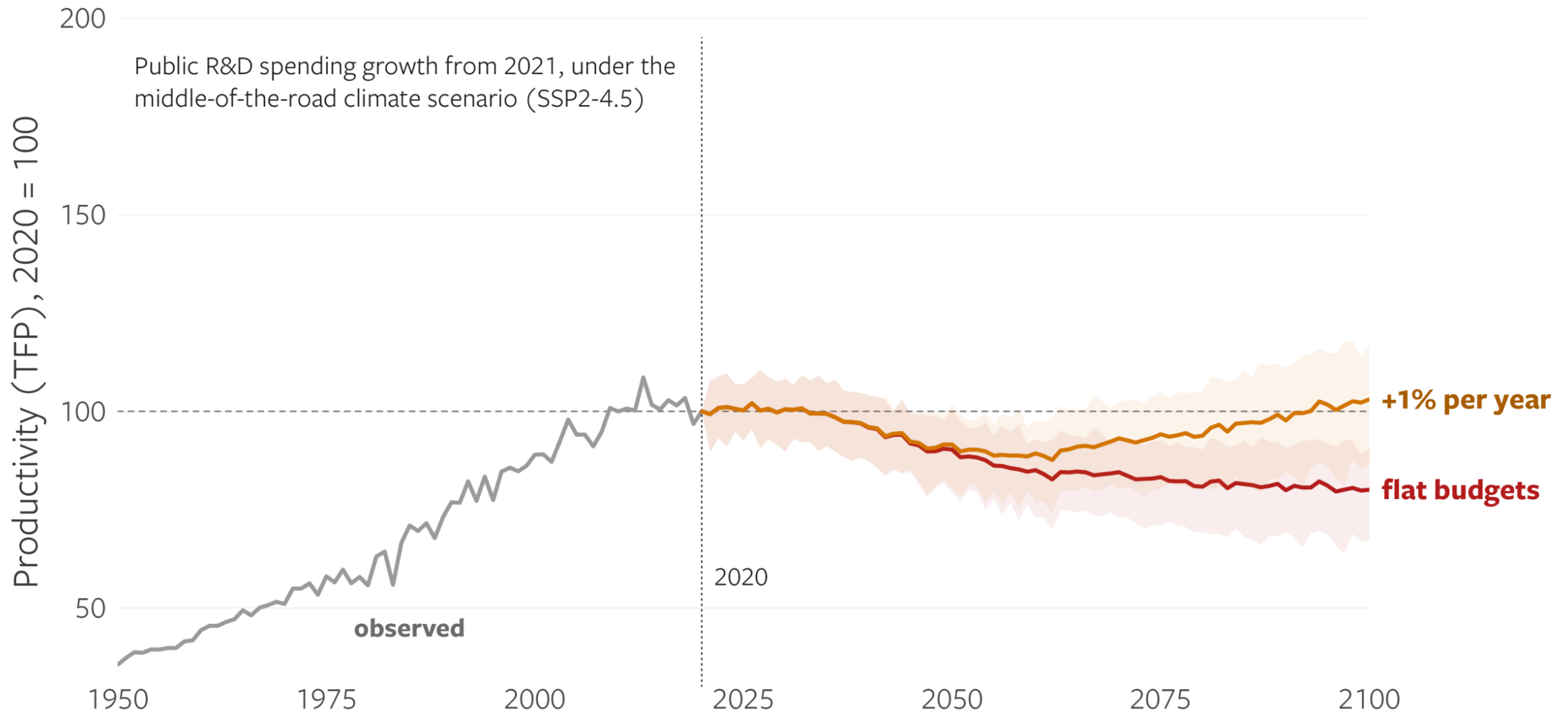
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4B and Table S12 (means and 95% CI).

Gradual budget growth means decades of delay before productivity recovers



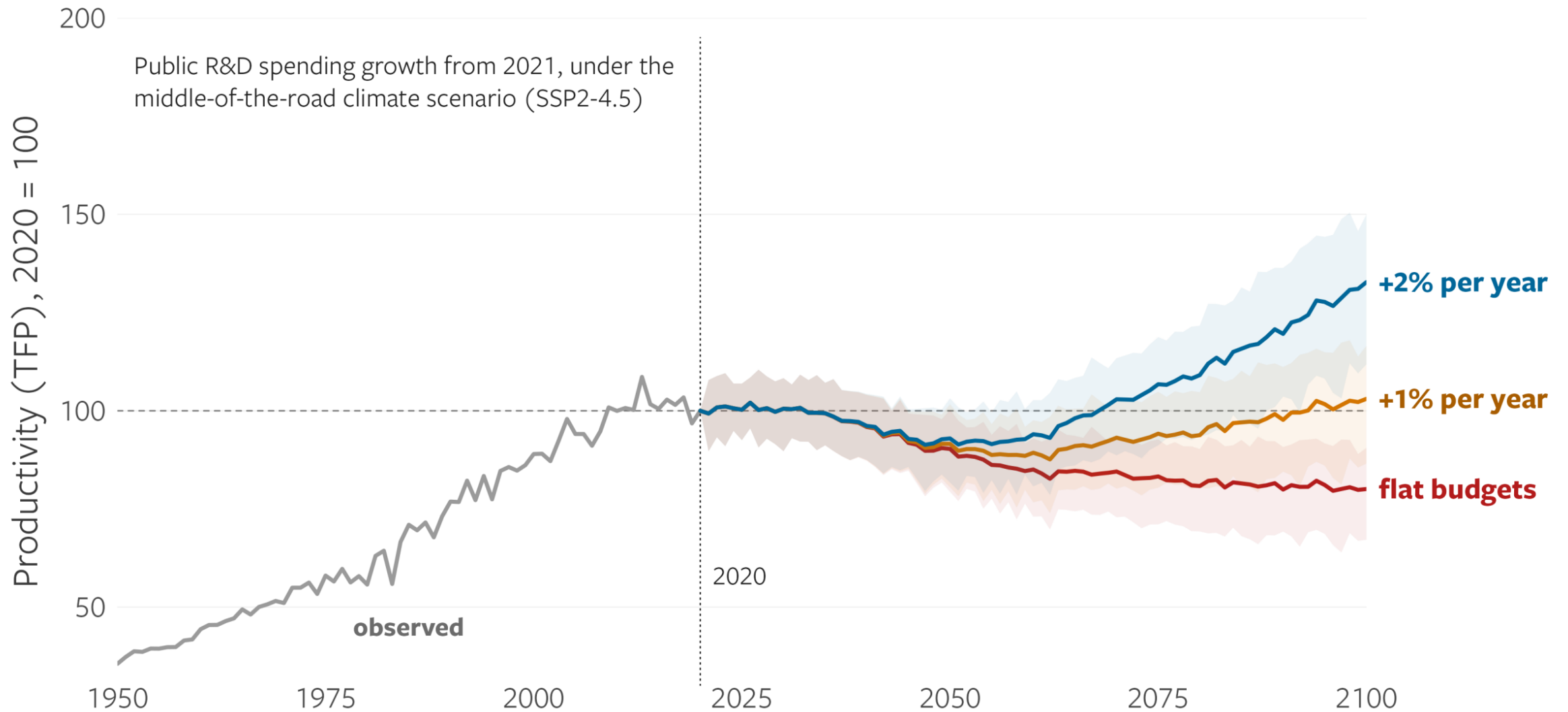
Source: Ortiz-Bobea et al. (PNAS 2025), projection following Fig. S26 (R&D lag structure, CMIP6 climate impacts); shaded = 90% range.

Gradual budget growth means decades of delay before productivity recovers



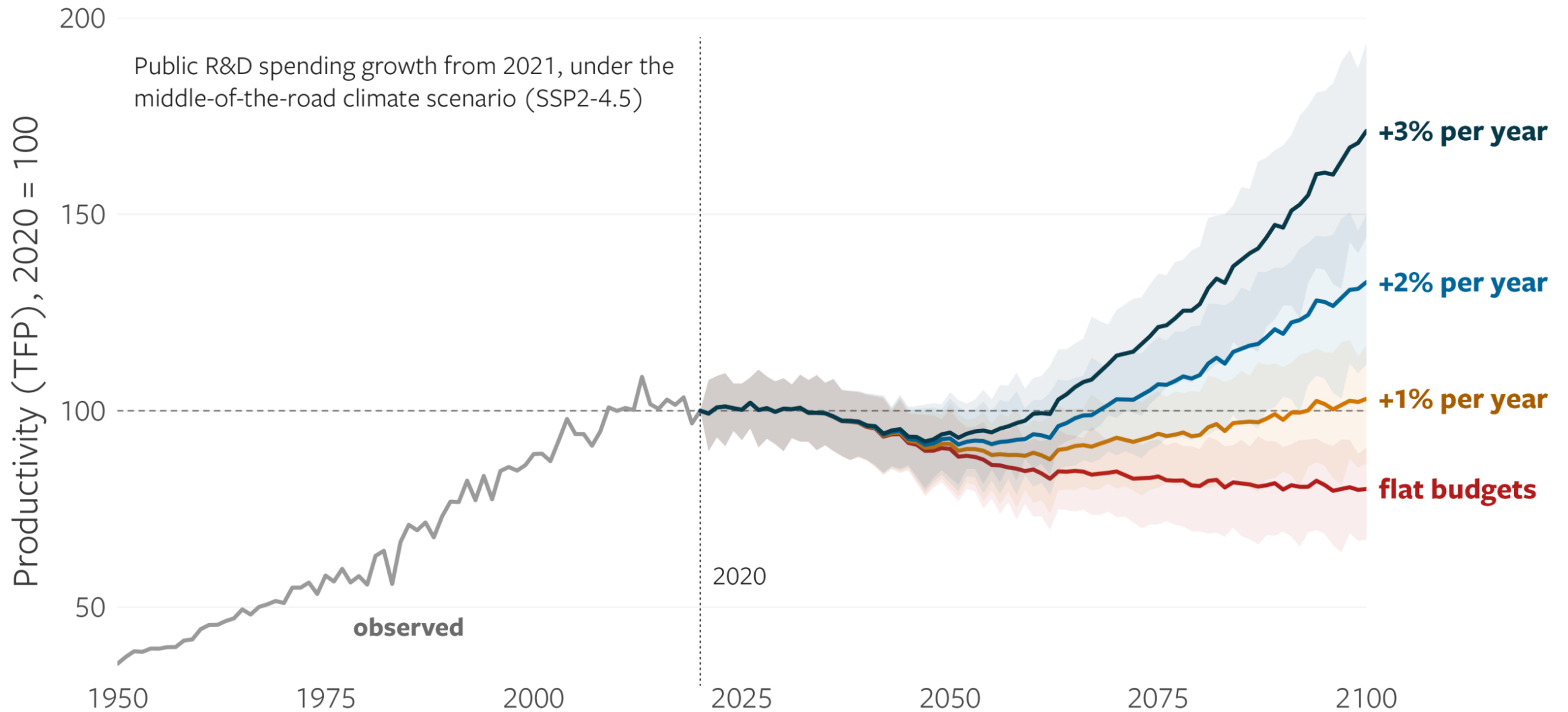
Source: Ortiz-Bobea et al. (PNAS 2025), projection following Fig. S26 (R&D lag structure, CMIP6 climate impacts); shaded = 90% range.

Gradual budget growth means decades of delay before productivity recovers



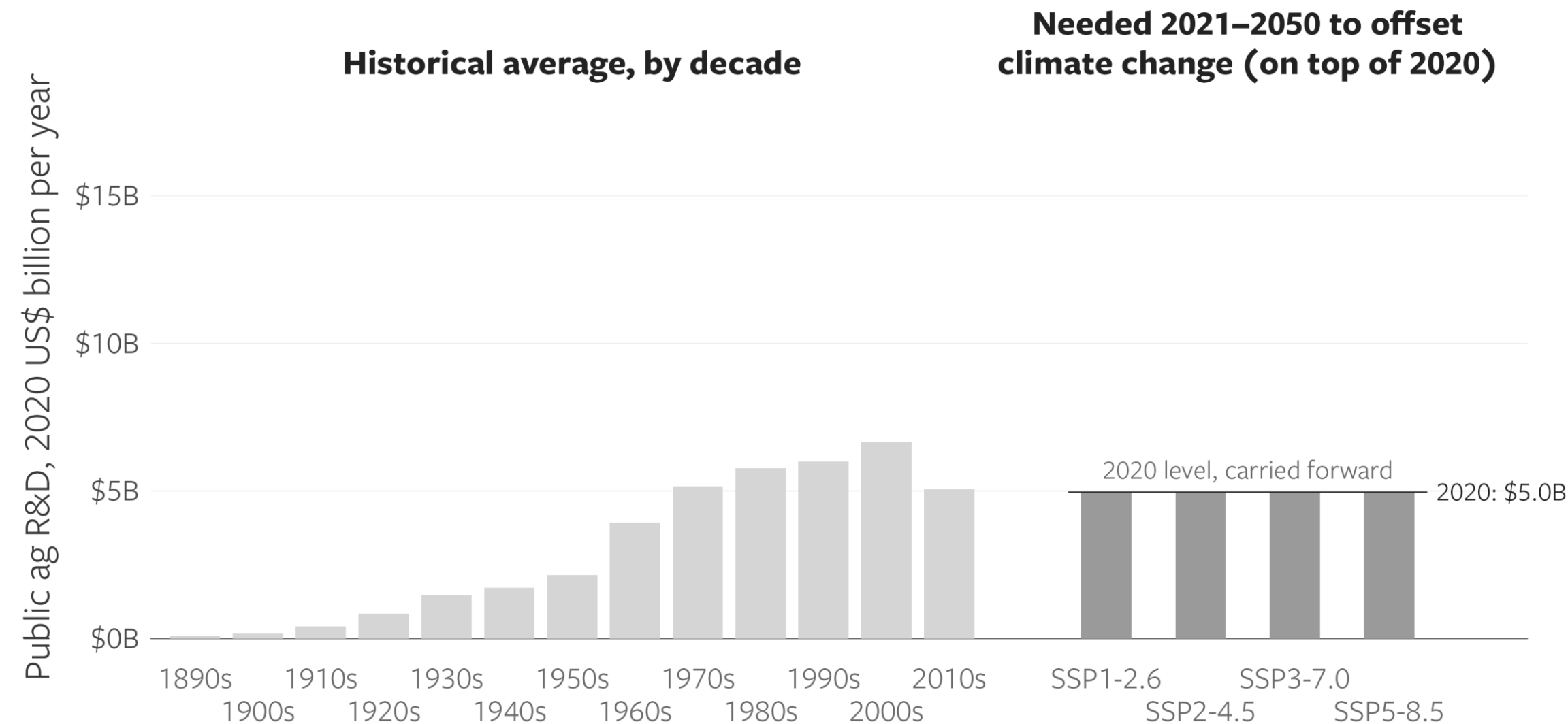
Source: Ortiz-Bobea et al. (PNAS 2025), projection following Fig. S26 (R&D lag structure, CMIP6 climate impacts); shaded = 90% range.

Gradual budget growth means decades of delay before productivity recovers



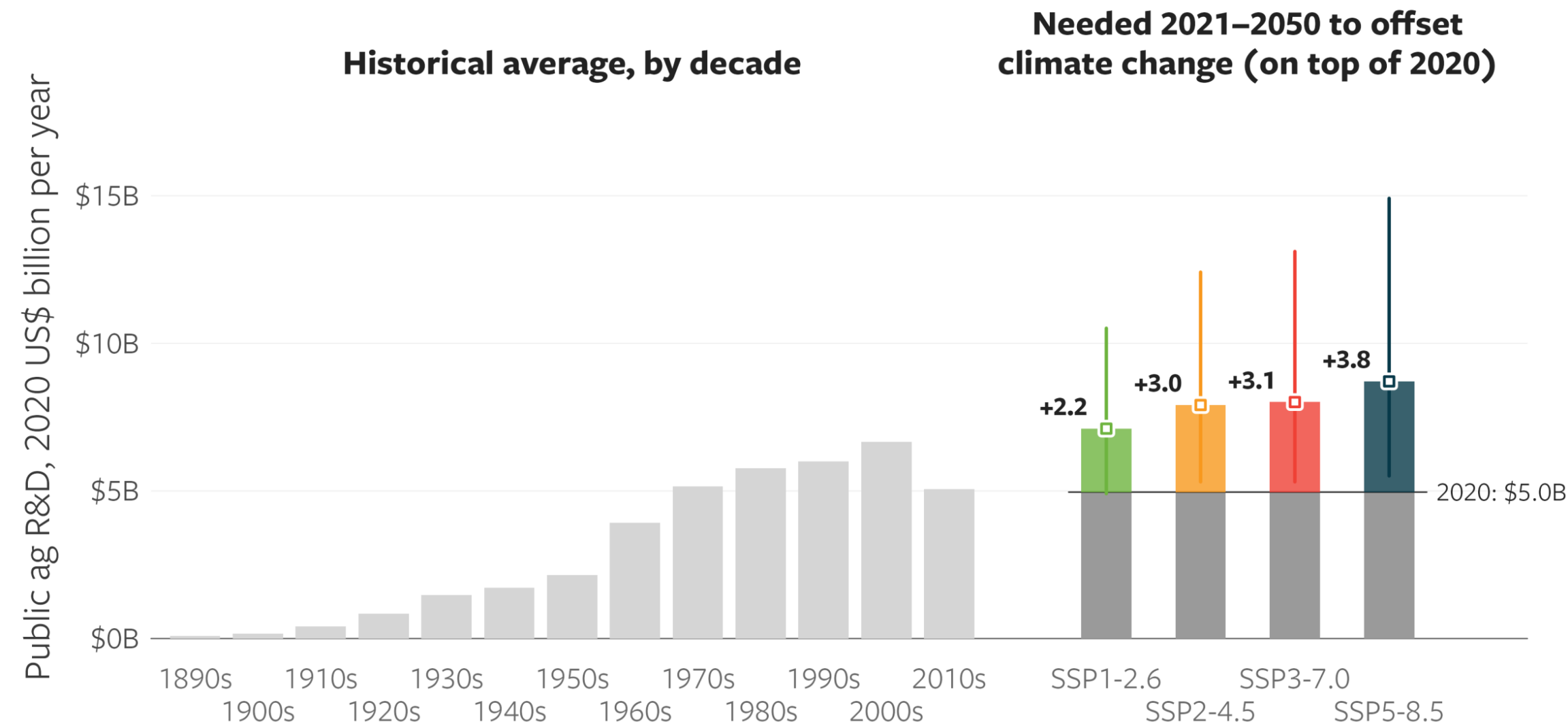
Source: Ortiz-Bobea et al. (PNAS 2025), projection following Fig. S26 (R&D lag structure, CMIP6 climate impacts); shaded = 90% range.

Strategy 2: a fixed supplement on top of today's \$5B



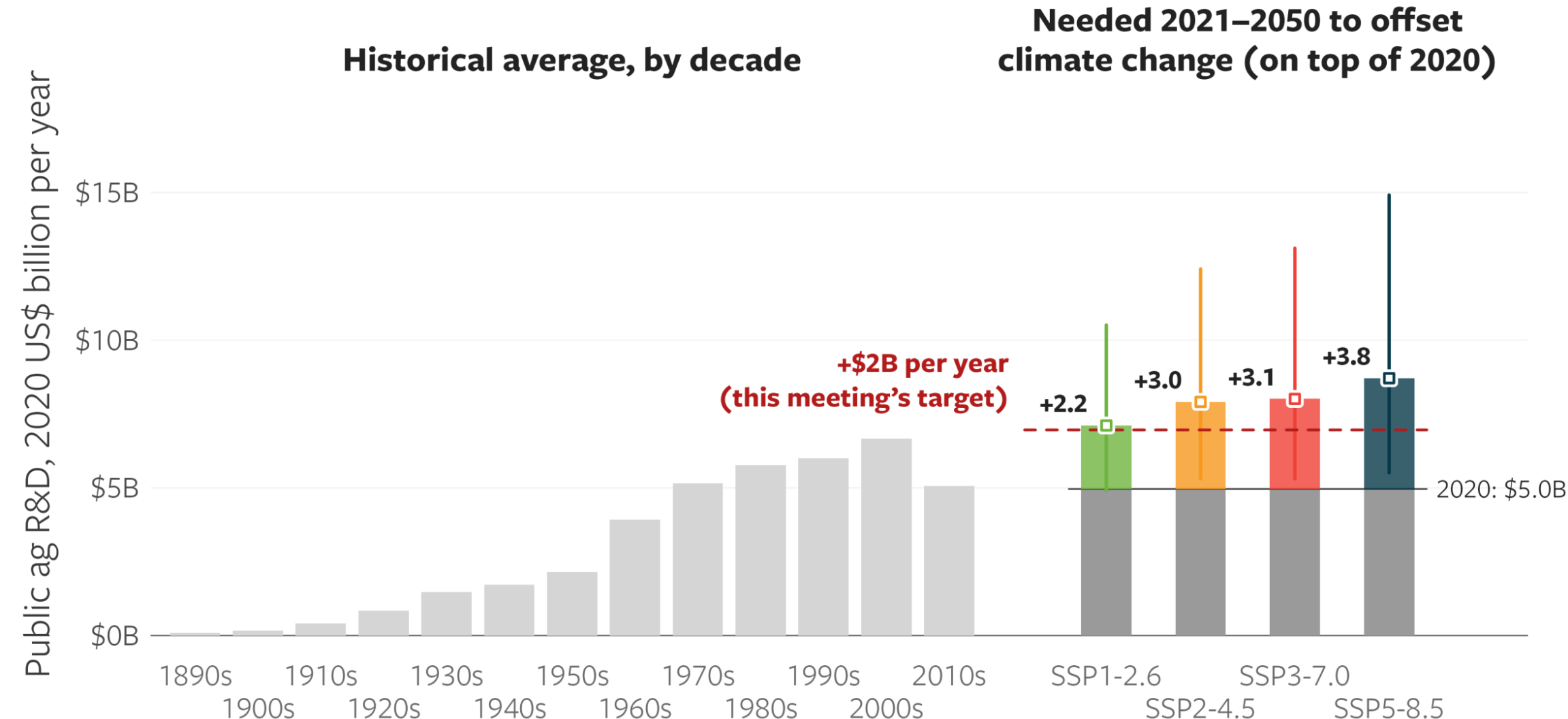
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14–S15 (fixed-supplement scenario, means and 95% CI).

It takes \$2–4B more per year just to offset climate



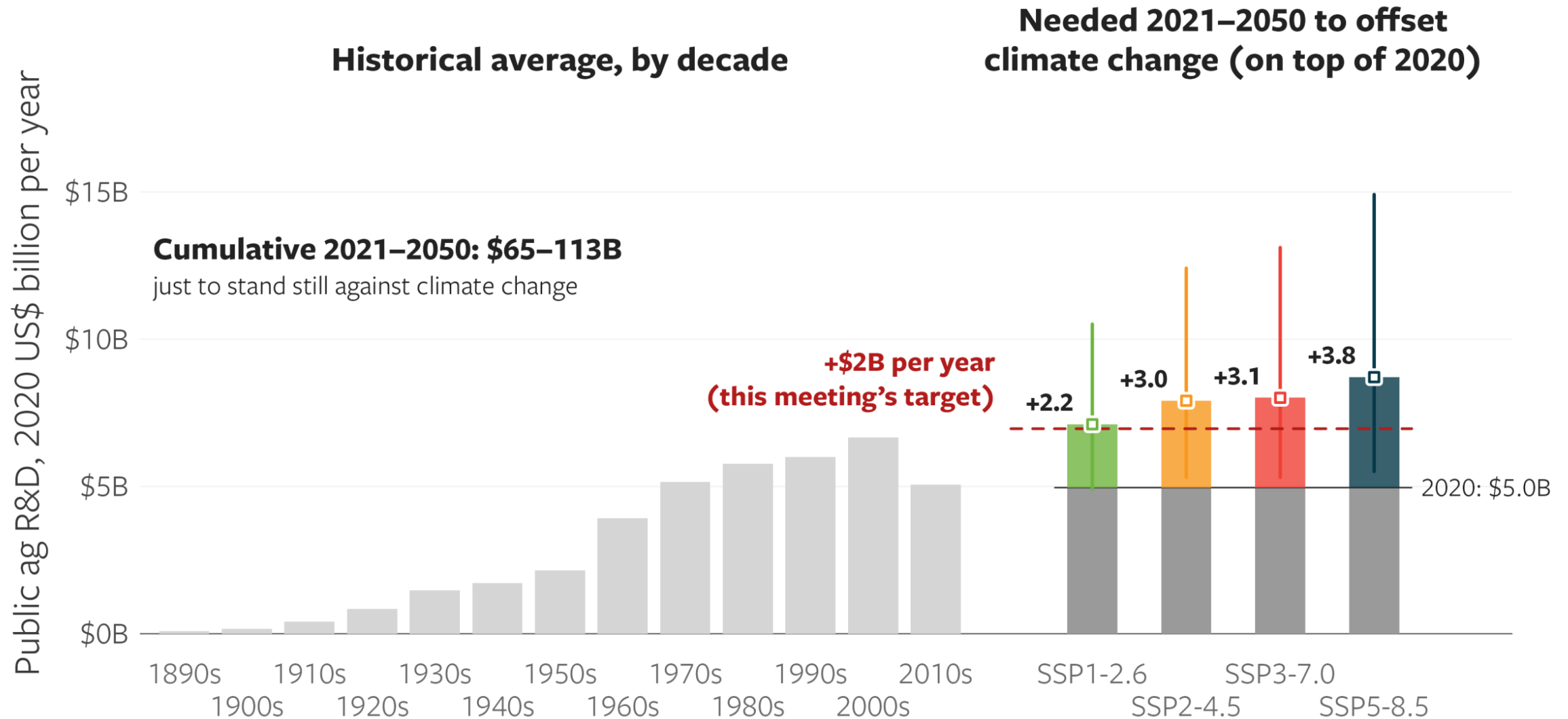
Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14–S15 (fixed-supplement scenario, means and 95% CI).

\$2B a year is the floor for standing still



Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14-S15 (fixed-supplement scenario, means and 95% CI).

\$65–113B by 2050, just to stand still



Source: USDA ERS spending (2020 US\$); Ortiz-Bobea et al. (PNAS 2025), Fig. 4C and Tables S14–S15 (fixed-supplement scenario, means and 95% CI).

What this number is not

No targeted adaptation research

We price the historical mix of research; adaptation-focused R&D could be cheaper — conservative.

Future returns like the past

We assume research pays off as it did over 1950–2020 — could cut either way.

Public R&D only

Private R&D is a complement, not a substitute; it is not in this number.

Key messages of this talk

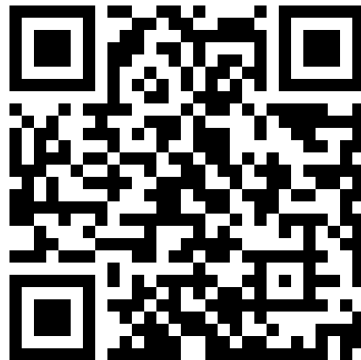
- 1 Productivity growth is slowing.**
- 2 The main driver is slowing R&D — but increasingly it's climate.**
- 3 Maintaining productivity growth requires a historical pivot.**

Thank you — let's keep in touch

Ariel Ortiz-Bobea

arielortizbobea.github.io

ao332@cornell.edu



the paper



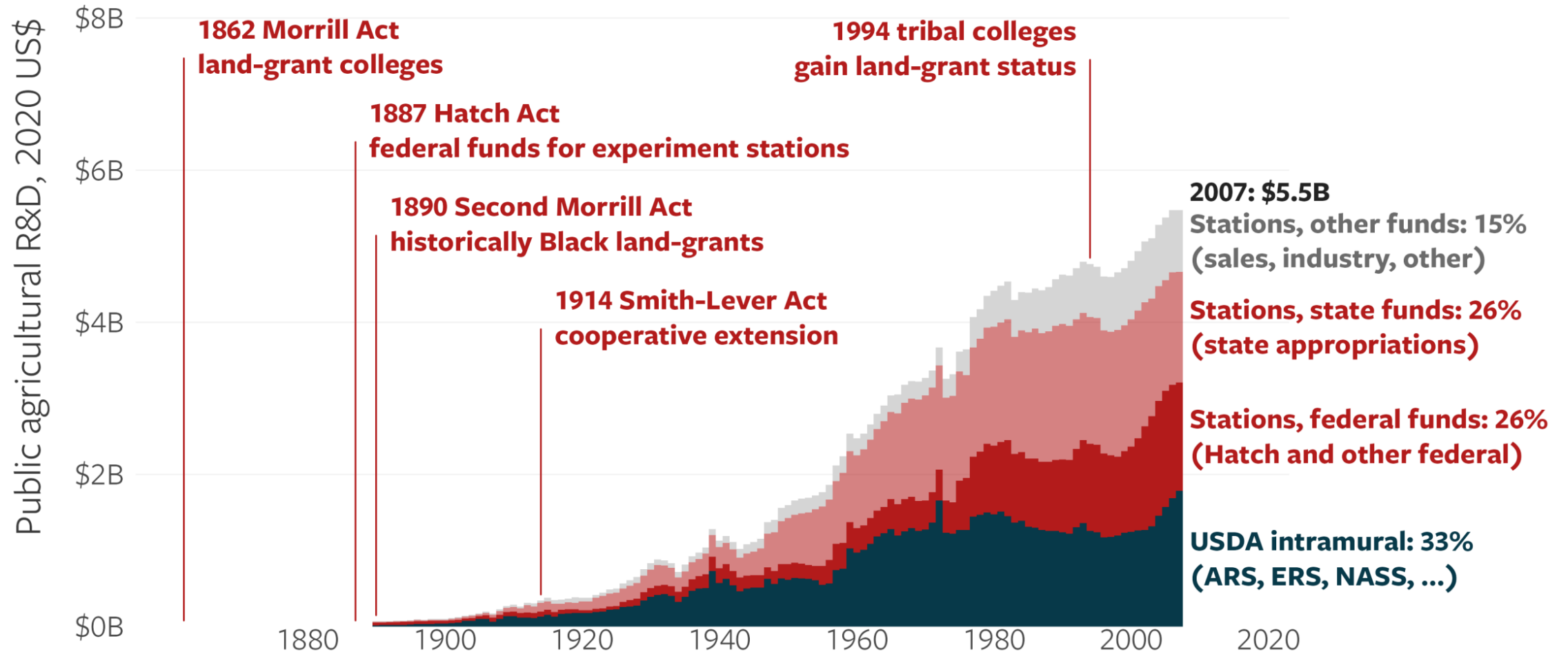
these slides

Extra slides

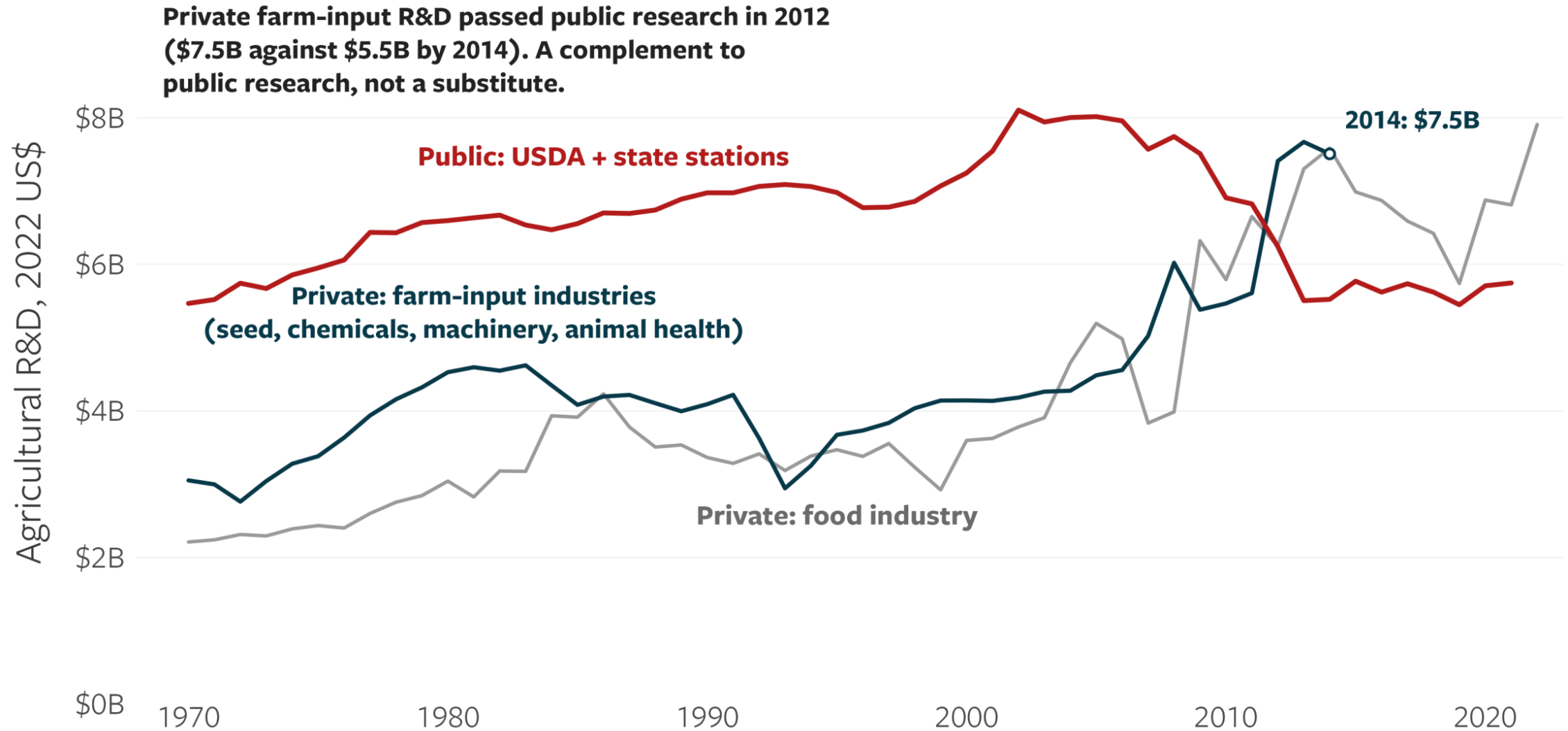
who pays for public research · public vs private R&D

Who spends it, and who pays: USDA and the state stations, by source of funds

InSTePP accounts (Pardey et al.), 1890–2007: USDA intramural + experiment stations, by source of funds; excl. forestry and extension; levels differ slightly from the ERS series (deflator, vet schools).



Private R&D has overtaken public research — and it is a complement, not a substitute



Source: USDA ERS, Agricultural and Food R&D Expenditures in the United States (1970–2022), constant 2022 US\$; private input-industry series ends in 2014.